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Simulating cosmic perturbations and the CMB power spectrum from inflationary initial conditions in ΛCDM

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ABSTRACT

We build a numerical Einstein-Boltzmann code from scratch, heavily inspired by the guide written by Callin (2006) and the lectures by Winther (2026a). The program simulates cosmic structure formation in a flat ΛCDM universe and computes the CMB and matter power spectra from first principles. The code was developped as a project in the course AST5220 at the University of Oslo during the Spring semester 2026. It solves the coupled Boltzmann hierarchies for cold dark matter, baryons, photons, and massless neutrinos from adiabatic inflationary initial conditions, using a tight-coupling approximation to stabilise the integration through the stiff pre-recombination epoch and the line-of-sight integral of Seljak & Zaldarriaga (1996) to recover the photon transfer functions at high multipoles. The recombination history is computed through the Saha and Peebles regimes with helium and reionisation included, giving a sound horizon at decoupling of $r_s \simeq 144.413$ Mpc. The predicted CMB temperature, E-mode polarisation, and cross power spectra, as well as the matter power spectrum, are found to be in excellent agreement with observations from Planck 2018 (Aghanim et al. 2020), SDSS, and the Lyα forest observations.

Key words. cosmology – cosmic microwave background – perturbation theory – Einstein-Boltzmann solver

0. Introduction

The cosmic microwave background (CMB) is the radiation released at recombination, when the Universe had cooled sufficiently for electrons and protons to combine into neutral hydrogen and the photons decoupled from the baryon plasma. The resulting temperature field is nearly uniform at $T_{\mathrm{CMB0}} \simeq 2.7255$ K today, but features tiny fluctuations of order $\delta T/T \sim 10^{-5}$, visible in the WMAP full-sky map in Figure 1. These perturbations encode the initial conditions set by inflation and the subsequent evolution of the photon-baryon plasma, making the CMB angular power spectrum one of the most powerful probes of cosmological parameters.

The large-scale structure of the Universe shares the same origin. Primordial perturbations seeded during inflation were amplified by gravity, and the acoustic oscillations of the photon-baryon plasma left coherent imprints on both the CMB and the matter distribution. Computing these observables from first principles therefore requires evolving a set of coupled perturbation equations for all species from deep in the radiation-dominated era to the present day.

In this project we build a numerical Einstein-Boltzmann code heavily based on the guide written by Callin (2006) and the curriculum in the course AST5220 teached by Hans A. Winther at the University of Oslo (Winther 2026a). The model simulates cosmic structure formation in a flat ΛCDM universe with Planck 2018 fiducial parameters (Aghanim et al. 2020), including cold dark matter, baryons, photons, massless neutrinos, and a cosmological constant. We first establish the background cosmology, then compute the recombination history, before evolving the full perturbation hierarchy using the tight-coupling approximation and the line-of-sight integral of Seljak & Zaldarriaga (1996) to efficiently recover the high-ℓ multipoles. Finally, we compute the CMB temperature, polarisation, and cross power spectra as well as the matter power spectrum, and compare our predictions to observational data.

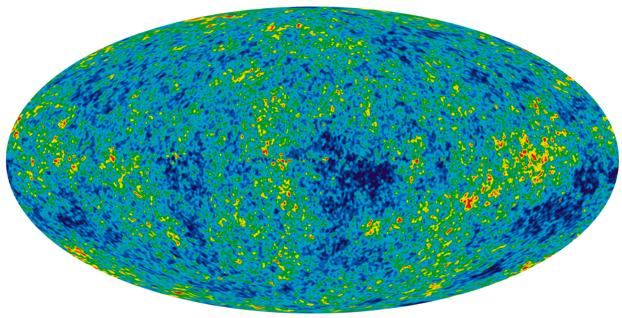

Fig. 1. CMB map (NASA WMAP Science Team 2025).

1. Background cosmology

To compute the CMB power spectrum, we must first establish the background cosmological model upon which perturbations will evolve. This section details the construction of this background Universe. We begin by outlining the theoretical framework, most notably the Friedmann-Lemaître-Robertson-Walker (FLRW) metric and the resulting Friedmann equations that governs the expansion history and the evolution of its energy components. Subsequently, we describe the numerical implementation of this model, including the fiducial parameters adopted from Planck 2018. Finally, we validate our implementation by comparing its predicted timeline of cosmic epochs and derived quantities, such as the age of the universe, against theoretical expectations, and we present the results of an MCMC parameter fit

⋆ The code developped for this project is available at GitHub.

to supernova data, which contextualises the well-known Hubble tension within our model.

1.1. Theory

In this section, we will present the relevant theory for developping the background cosmology in our model Universe along with all of the equations used. All of the theory is covered in greater detail in the lecture notes by Hans A. Winther for the cosmology course AST5220 (Winther 2026a,b), so we will simply summarise the key details and derive some quantities that will come in handy for the numerical implementation.

1.1.1. The Friedmann-Lemaître-Robertson-Walker geometry

For our background cosmology, we assume an isotropic and homogeneous Universe with a spatial scale factor $a(t)$ equipped with the Friedmann-Lemaître-Robertson-Walker (FLRW) metric

$$\mathrm{d}s^2 = -\mathrm{d}t^2 + a^2(t)\Big(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\Big), \tag{1}$$

or, on component form; $g_{00} = -1$, $g_{ij} = a^2(t)\delta_{ij}$.

At the background level, the constituents of our Universe, which include radiation (photons and massless neutrinos), non-relativistic matter (baryons and cold dark matter), and a cosmological constant, are treated as perfect fluids. For each component i, the stress-energy tensor in the comoving frame takes the diagonal form

$$T^{(i)}_{\mu\nu} = \mathrm{diag}(\rho_i, p_i, p_i, p_i),$$

with ρ_i denoting the density and p_i the pressure of that component.

The resulting Einstein field equations for the FLRW metric and the sum of the stress-energy tensors for the perfect fluids,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G \sum_i T^{(i)}_{\mu\nu}, \tag{2}$$

we get the Friedmann equations. The (00) component of the Einstein equations gives us the first equation,

$$H^2(a) \equiv \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\sum_i \rho_i, \tag{3}$$

while the (11), (22) and (33) components gives us the second Friedmann equation (assuming isotropy, or else they would yield three different equations),

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\sum_i (\rho_i + 3p_i). \tag{4}$$

As our time variable, we will use the logarithm of scale factor $x \equiv \ln[a(t)]$. We thus need to study the Friedmann equations, Eqs.(3)-(4), to understand how the scale factor evolves and how it is related to the evolution of the densities of the different components in our model Universe.

1.1.2. The Hubble function

Introducing the definition of the density parameters $\Omega_i \equiv 8\pi G\rho_i/3H^2$, and the equation of state $w_i \equiv p_i/\rho_i$, the first Friedmann equation can be recast into the form we will be using for our implementation of the background cosmology

$$H^2 = H_0^2 \sum_i \frac{\Omega_{i0}}{a^{3(1+w_i)}},$$

where $H_0 = 100h$ km/s/Mpc is the Hubble constant, and Ω_{i0} is the current day value of the density parameter of component i.

In our model Universe, we have dark energy, described by a cosmological constant (Λ) with equation of state $w_\Lambda = -1$, non-relativistic matter in the form of baryons (b) and cold dark matter (CDM) with equation of state $w_M = 0$, and two relativistic species, photons (γ) and neutrinos (ν) with $w_R = 1/3$. The curvature (k) can be decribed as having an equation of state $w_k = -1/3$. We can thus explicitly write out our Hubble function as

$$H(a) = H_0\sqrt{\Omega_{\Lambda 0} + \frac{\Omega_{k0}}{a^2} + \frac{\Omega_{b0} + \Omega_{\mathrm{CDM}0}}{a^3} + \frac{\Omega_{\gamma 0} + \Omega_{\nu 0}}{a^4}}. \tag{5}$$

The evolution of the density parameters as functions of the scale factor are thus given by

$$\Omega_k(a) = \frac{\Omega_{K0}}{a^2 H(a)^2/H_0^2}, \qquad \Omega_M(a) = \frac{\Omega_{M0}}{a^3 H(a)^2/H_0^2},$$
$$\Omega_R(a) = \frac{\Omega_{R0}}{a^4 H(a)^2/H_0^2}, \qquad \Omega_\Lambda(a) = \frac{\Omega_{\Lambda 0}}{H(a)^2/H_0^2}. \tag{6}$$

Here we define the non-relativistic matter density parameter $\Omega_M = \Omega_b + \Omega_{\mathrm{CDM}}$ and the relativistic species density parameter $\Omega_R = \Omega_\gamma + \Omega_\nu$.

1.1.3. Cosmic and conformal time

In the FLRW metric, Eq.(1), we use the *cosmic time* coordinate t which is related to the scale factor through the Hubble function through the relation

$$H \equiv \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}t} \;\Rightarrow\; \frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{H},$$

or on integral form

$$t(x) = \int_{-\infty}^{x} \frac{\mathrm{d}x'}{H(x')}. \tag{7}$$

However, we also consider the *conformal time* η which is the distance that light may have travelled since the Big Bang (horizon). It is related to the cosmic time through the relation

$$a\mathrm{d}\eta = c\mathrm{d}t \;\Rightarrow\; \frac{\mathrm{d}\eta}{\mathrm{d}x} = \frac{c}{aH}.$$

For calculating η, it is also prudent to introduce the conformal Hubble function $\mathcal{H} \equiv \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}\eta} = aH$ such that the conformal time can be written on integral form as

$$\eta(x) = \int_{-\infty}^{x} \frac{c\,\mathrm{d}x'}{\mathcal{H}(x')}. \tag{8}$$

The derivatives of $\mathcal{H}$ with respect to x are given by;

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} = \mathcal{H}\left[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\right]\Bigg|_{a=e^x}, \tag{9}$$

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} = \;& \mathcal{H}\Bigg\{\left[1 - \frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\right]^2 \\ & - \frac{1}{2}\Big(3\Omega_M + 4\Omega_R + 2\Omega_K\Big)^2 + \frac{1}{2}\Big(9\Omega_M + 16\Omega_R + 4\Omega_K\Big)\Bigg\}\Bigg|_{a=e^x},\end{aligned} \tag{10}$$

where Ω_i are the density parameters evaluated at $a = e^x$. The derivatives of $\mathcal{H}$ are derived in Appendix A.

1.1.4. Luminosity distance and angular diameter distance

In order to compare our model with observations, and be able to perform parameter fitting, we will need to calculate two observables; namely the *angular diameter distance* d_A, and the *luminosity distance* d_L.

Writing the line element in the FLRW metric, Eq.(1), using spherical coordinates, we have

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + a^2\left(\frac{\mathrm{d}r^2}{1-kr^2} + r^2\mathrm{d}\theta^2 + r^2\sin\theta^2\mathrm{d}\phi^2\right), \tag{11}$$

where k is the curvature. Since we are observing photons (that move radially along the line of sight) when determining astrophysical distances, we consider the null geodesic, $\mathrm{d}s^2 = 0$, followed by the rays of light. For these photons, we get

$$c\mathrm{d}t = \frac{a\mathrm{d}r}{\sqrt{1-kr}} \;\Rightarrow\; \chi \equiv \eta_0 - \eta = \int_0^r \frac{\mathrm{d}r'}{\sqrt{1-kr'}}. \tag{12}$$

Here, χ denotes the *comoving distance*.

Solving the integral in Eq.(12) and isolating r, we find three solutions, depending on the curvature

$$r = \begin{cases} \dfrac{\chi\sin\left[\sqrt{|\Omega_{k0}|}H_0\chi/c\right]}{\sqrt{|\Omega_{k0}|}H_0\chi/c} & \Omega_{k0} < 0, \\ \chi & \Omega_{k0} = 0, \\ \dfrac{\chi\sinh\left[\sqrt{|\Omega_{k0}|}H_0\chi/c\right]}{\sqrt{|\Omega_{k0}|}H_0\chi/c} & \Omega_{k0} > 0. \end{cases}$$

If an object has physical size D and subtends an angle θ on the sky, the angular diameter distance is defined as $d_A = D/\theta$. Reading off the line element, Eq.(11), a displacement in the θ-direction corresponds to a proper distance $dD = ar\,d\theta$, so

$$d_A = D/\theta = ar.$$

On the other hand, from the intrinsic luminosity of a source and its observed flux, the luminosity distance d_L can be defined through $F = L/4\pi d_L^2$, which simply reads

$$d_L = a/r = d_A/a^2.$$

1.1.5. Some cosmological events

Finally, for our analysis of the simulated data, it is convenient to identify key cosmological events to serve as heuristic markers for our interpretation of the results. We will look at three relevant events at the time scale of the evolution of the background:

(i) matter-radiation equality at $x_{\mathrm{eq}}^{MR} = \ln\left(\frac{\Omega_{R0}}{\Omega_{M0}}\right)$;

(ii) the onset of cosmic acceleration (when $\ddot{a} = 0$) at

$$x_{\mathrm{acc}} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{2\Omega_{\Lambda 0}}\right);$$

(iii) and the matter-dark energy equality at

$$x_{\mathrm{eq}}^{\Lambda M} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{\Omega_{\Lambda 0}}\right).$$

For the derivations of the three x's, see Appendix B.

1.2. *Implementation details*

1.2.1. Fiducial cosmological parameters

Our model Universe will assume ΛCDM fiducial cosmology and be initialised with the best-fit cosmological parameters from the Planck 2018 data (Aghanim et al. 2020). We will thus take the dimensionless Hubble parameter $h = 0.67$, the current day temperature of the CMB radiation $T_{\mathrm{CMB0}} = 2.7255$ K, and the effective number of massless neutrino species $N_{\mathrm{eff}} = 3.046$.

For the current day values of the density parameters, we set

$$\begin{aligned} \Omega_{b0} &= 0.05, \\ \Omega_{\mathrm{CDM0}} &= 0.267, \\ \Omega_{K0} &= 0, \\ \Omega_{\gamma 0} &= \frac{\pi^2}{15}\frac{(k_b T_{\mathrm{CMB0}})^4}{\hbar^3 c^5}\frac{8\pi G}{3H_0^2} \simeq 5.509\times 10^{-5}, \\ \Omega_{\nu 0} &= \frac{7}{8}\left(\frac{4}{11}\right)^{4/3} N_{\mathrm{eff}}\Omega_{\gamma 0} \simeq 3.811\times 10^{-5}, \\ \Omega_{\Lambda 0} &= 1 - \sum_{i\neq\Lambda}\Omega_{i0} \simeq 0.683. \end{aligned}$$

Here, the dark energy density parameter is defined such that $\sum_i \Omega_{i0} = 1$, to ensure a spatially flat Universe.

1.2.2. Numerical integration scheme

To evaluate integral quantities such as Eqs.(7) and (8), we recast them as ordinary differential equations and integrate numerically using the fourth-order Runge-Kutta solver provided by the GSL library (GNU Project 2024) and spline the result using the cubic spline algorithm available in GSL.

The cosmic time ODE is

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{1}{H(x)},$$

which is to be integrated up from $x_i = -\infty$ to some final x_f. However, in practice, we must choose a very early time to integrate from. In our case, we solve in the interval $x_i = -20$ up to $x_f = 5$. The initial condition for the ODE is determined deep inside the radiation dominated era, when $H(x) \sim H_0\sqrt{\Omega_{R0}}e^{-2x}$, such that

$$\frac{\mathrm{d}t}{\mathrm{d}x} = \frac{e^{2x}}{H_0\sqrt{\Omega_{R0}}} \;\Rightarrow\; t(x_i) = \int_{-\infty}^{x_i}\frac{e^{2x}\mathrm{d}x}{H_0\sqrt{\Omega_{R0}}} = \frac{1}{2H(x_i)}.$$

Similarly, we solve the ODE for the conformal time

$$\frac{\mathrm{d}\eta}{\mathrm{d}x} = \frac{c}{\mathcal{H}(x)}$$

in the same interval, with the initial condition $\eta(x_i) = c/\mathcal{H}(x_i)$, due to the radiation dominated conformal Hubble function $\mathcal{H}(x) \sim H_0\sqrt{\Omega_{R0}}e^{-x}$.

1.2.3. MCMC parameter fitting

Using Markov Chain Monte Carlo (MCMC), we will test our simulated Universe, based on fiducial cosmological parameters, against observations of type Ia supernova (SNIa) luminosity distances (Betoule et al. 2014). The procedure searches for the triplet of "best-fit" parameters in the $(h, \Omega_{M0}, \Omega_{K0})$ parameter space that minimises

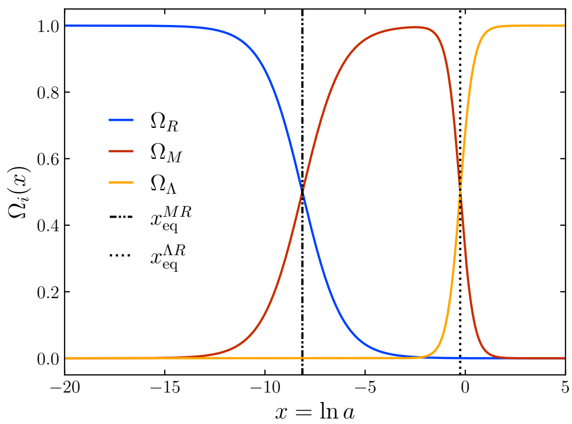

Fig. 2. Evolution of the density parameters of relativistic species (Ω_R), non-relativistic matter (Ω_M), and of dark energy (Ω_Λ) as functions of $x = \ln a$ on the interval $-20 \leq x \leq 5$. Plotted with vertical dashed and dotted lines are also the times of matter-radiation and matter-dark energy equality.

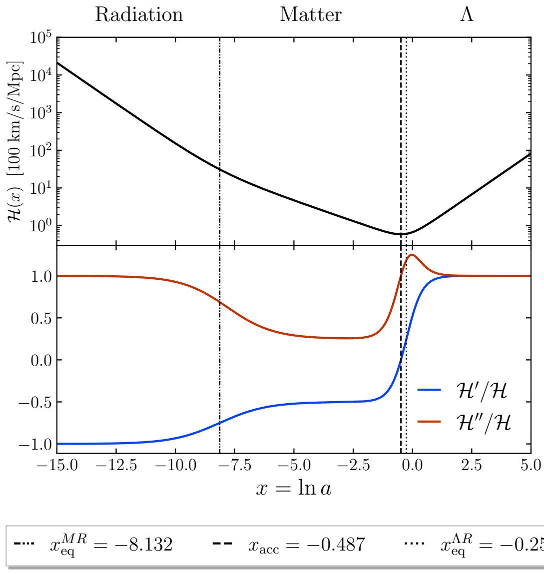

Fig. 3. The conformal Hubble function $\mathcal{H}(x)$ along with $\mathcal{H}'/\mathcal{H}$ and $\mathcal{H}''/\mathcal{H}$ plotted as functions of x. The time at which we have the onset of cosmic acceleration $x_{\rm acc}$, and the equality times $x_{\rm eq}^{mr}$ and $x_{\rm eq}^{\Lambda m}$ are shown with vertical dotted lines with the dominating component indicated above the regions in between them.

$$\chi^2(h, \Omega_{M0}, \Omega_{K0}) = \sum_{i=1}^{N} \frac{[d_L(z_i, h, \Omega_{M0}, \Omega_{K0}) - d_L^{\rm obs}(z_i)]^2}{\sigma_i^2}.$$

Here, d_L are the predicted luminosity distances from our model at the observed redshifts z_i, while $d_L^{\rm obs}$ are the observed distances, together with their associated errors.

To perform the actual MCMC parameter fitting procedure, we employ the code written Winther (2026f).

1.3. Results

1.3.1. Timeline of the background

In Table 1, we have stated the times for the matter-radiation equality, the onset of cosmic expansion, and the matter-dark energy equality, in terms of the log scale factor x, the redshift $z(x) = e^{-x} - 1$ and the cosmic time $t(x)$. We remark that cosmic acceleration only begins in the very late Universe at around $z \sim 0.6$. We thus observe the same late-stage acceleration as we would expect from the ΛCDM Universe.

Displayed in Figure 2 are the density parameters Ω_R, Ω_M, and Ω_Λ as functions of x, representing the contributions from relativistic components, non-relativistic matter, and the cosmological constant. The figure highlights three distinct periods in the history of our model Universe:

(i) an era of radiation domination in the very early Universe until $x_{\rm eq}^{MR} = -8.132$;

(ii) a matter dominated period between $x_{\rm eq}^{MR} = -8.132$ and $x_{\rm eq}^{\Lambda M} = -0.256$;

(iii) and a late stage dark energy dominated Universe after $x_{\rm eq}^{\Lambda M} = -0.256$.

This timeline is explained by the fact that $\rho_R \propto a^{-4}$, $\rho_M \propto a^{-3}$, while $\rho_\Lambda = $ const, such that the density of the relativistic components decays faster than non-relativistic matter, which in turn decays faster than the density of dark energy.

The transition between cosmological eras leaves clear signatures in the conformal Hubble function and its derivatives, as illustrated in Figure 3. Equations (9) and (10) predict specific values in each regime: $\mathcal{H}'/\mathcal{H} = -1$ and $\mathcal{H}''/\mathcal{H} = 1$ in radiation domination, $\mathcal{H}'/\mathcal{H} = -1/2$ and $\mathcal{H}''/\mathcal{H} = 1/4$ in matter domination, and both ratios approaching unity in a Λ-dominated universe. These predicted features are apparent in the bottom panel of Figure 3.

The top panel also exhibits the cosmic acceleration after $x_{\rm acc}$. Since $\mathcal{H} \equiv aH = \dot{a}$, the conformal Hubble function describes the rate of change of the scale factor with respect to cosmic time. We observe that the slope of $\mathcal{H}$ changes from negative to positive, i.e. $\mathcal{H}'/\mathcal{H}$ becomes positive, at the minimum situated at $x = x_{acc}$. A positive slope means that

$$\mathcal{H}' = \frac{{\rm d}\dot{a}}{{\rm d}x} = \frac{{\rm d}t}{{\rm d}x}\ddot{a} > 0.$$

Since $t(x)$ is manifestly a strictly increasing function, ${\rm d}t/{\rm d}x$ is strictly positive at all times, such that the sign of $\mathcal{H}'$ depends on the sign of $\ddot{a}$. Therefore, we see that $\ddot{a} > 0$ for $x > x_{\rm acc}$, so we indeed have late-stage cosmic acceleration.

1.3.2. Cosmic and conformal time, and the age of the Universe

Figure 4 presents the conformal time $\eta(x)$ as well as the dimensionless quantity $\eta\mathcal{H}/c$ in the interval $-18 \leq x \leq 0$. From the initial condition $\eta(x_i) = c/\mathcal{H}(x_i)$, derived in Section 1.2.2, we expect that $\eta\mathcal{H}/c \to 1$ as $x \to -\infty$. We observe that this is in fact the case for our model (see top panel, Fig.4). The conformal time is also a smooth monotonically increasing function of x, and its current day value is calculated to be $\eta(0)/c \simeq 46.319$ Gyr.

The solution for the cosmic time $t(x)$ is showcased in Figure 5 over the interval $-8 \leq x \leq 5$, with a current day value determined to be $t_0 \equiv t(0) \simeq 13.858$ Gyr. During the matter domination era, the Hubble function is $H(x) \propto e^{-3x/2}$, so we would

Equality times and onset of acceleration			
Event	Log scale factor x	Redshift z	Time t [Gyr]
Matter-radiation equality (MR)	-8.132	3400.33	5.106×10^{-5}
Onset of acceleration (acc)	-0.487	0.6272	7.752
Matter-dark energy equality (ΛM)	-0.256	0.2915	10.378

Table 1. Times for when we have matter-radiation equality, onset of cosmic acceleration, and matter-dark energy equality expressed using x, redshift $z = e^{-x} - 1$, and cosmic time $t(x)$.

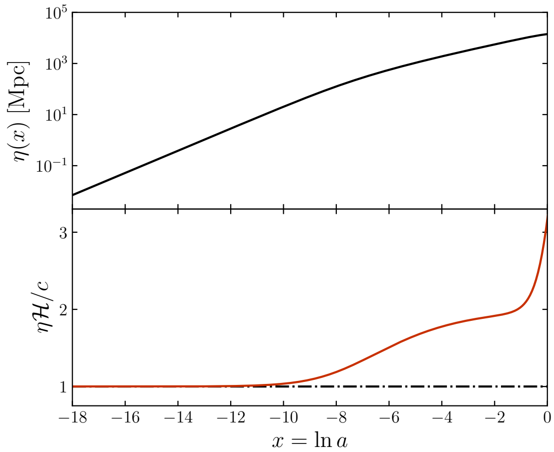

Fig. 4. The top panel presents the evolution of the conformal time with respect to x. The bottom panel shows the dimensionless quantity $\eta\mathcal{H}/c$ over the same interval.

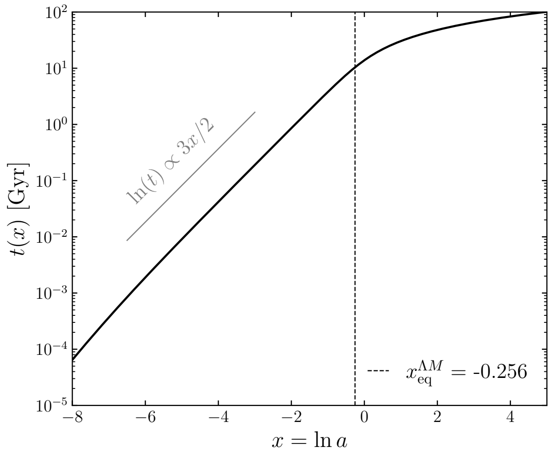

Fig. 5. Cosmic time with respect to x. The gray diagonal line is included in order to compare the slope of $t(x)$ with our expectation of $\ln t \propto 3x/2$ during matter domination.

expect that

$$\frac{\mathrm{d}t}{\mathrm{d}x} \propto e^{3x/2} \;\Rightarrow\; \ln t(x) \propto \frac{3}{2}x.$$

Comparing to the gray line included in Figure 5, we see that it matches the slope of $t(x)$ between $x = -8$ up to $x \sim x^{\Lambda M}_{\rm eq}$. In the dark energy dominated era ($x \gtrsim x^{\Lambda M}_{\rm eq}$), the slope decreases and

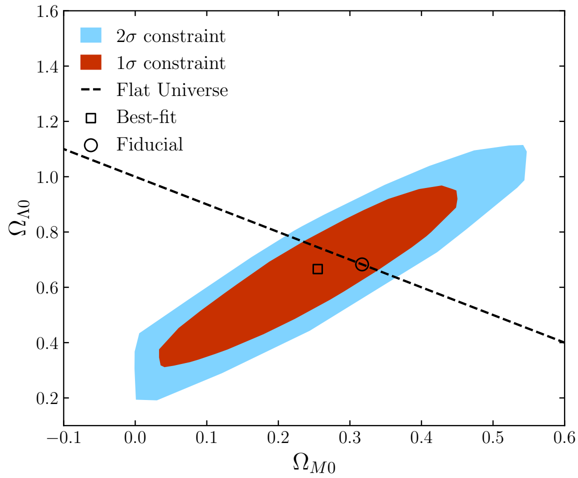

Fig. 6. The regions of sampled points in the $(\Omega_{M0}, \Omega_{\Lambda 0})$ parameter space within 1σ and 2σ of the pair associated with the minimal χ^2. The diagonal dashed line shows the cosmologies giving a flat Universe; $\Omega_{K0} = 1 - \Omega_{M0} - \Omega_{\Lambda 0} = 0$.

the curve flattens out. This is to be expected from the fact that $H \sim H_0\sqrt{\Omega_{\Lambda 0}} = $ const such that $t \propto x$ during this era.

1.3.3. MCMC best-fit parameters and the Hubble tension

The MCMC parameter fitting to SNIa luminosity distance data (see Sect.1.2.3) yielded the best-fit triplet of parameters

$$h = 0.701711,$$
$$\Omega_{M0} = 0.255027, \quad 280$$
$$\Omega_{K0} = 0.0789514,$$

with an associated $\chi^2 = 29.2799$. The observational data contains $N = 31$ data, so we have $\chi^2/N \lesssim 1$, indicating a good fit.

Figure 6 shows the $(\Omega_{M0}, \Omega_{\Lambda 0})$ parameter space sampled during the MCMC procedure, highlighting the regions within 1σ and 2σ of the best-fit parameters (minimum χ^2). The fiducial Planck values lie within the 1σ contour, though slightly offset from the best-fit point.

The marginalized posterior distributions for $\Omega_{\Lambda 0}$ and the Hubble parameter h are presented in Figure 7, with vertical dotted lines indicating the fiducial Planck values. While the fiducial $\Omega_{\Lambda 0}$ falls comfortably within the posterior and close to its best-fit value, the same is not true for h. The fiducial value of h lies well outside the 1σ region of its posterior distribution, with a relative difference of

$$\left|\frac{h_{\text{best-fit}} - h_{\text{fiducial}}}{h_{\text{best-fit}}}\right| \simeq 4.5\%.$$

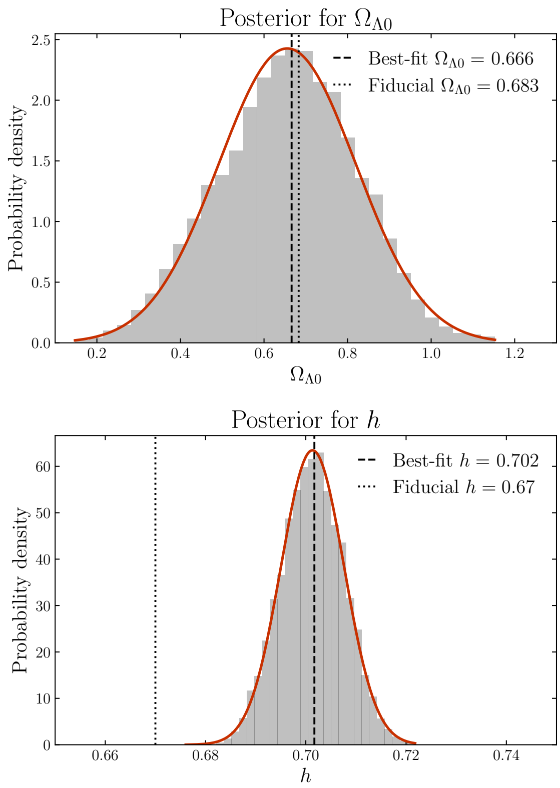

Fig. 7. Posteriors for $\Omega_{\Lambda 0}$ and h from the MCMC parameter fitting. In the posterior for h, we have also indicated the fiducial value of $h = 0.67$ with a blue vertical line to highlight the Hubble tension.

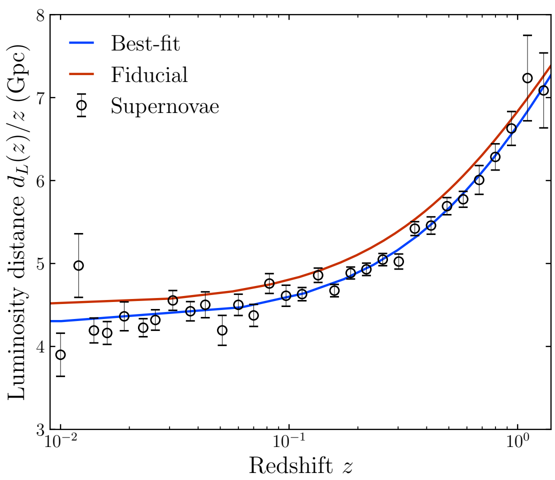

Fig. 8. Luminosity distance as a function of redshift z. The data points represent the luminosity distances deduced from observations of supernovae at different redshifts. The two curves show the luminosity distance functions $d_L(z)/z$ for the best fit parameters and for the fiducial cosmological parameters.

This discrepancy is further illustrated in Figure 8, which compares the luminosity distance-redshift relation predicted by our best-fit model to that of the fiducial Planck cosmology, alongside the observed SNIa data. The best-fit parameters clearly provide a superior match to the supernova measurements across the full redshift range, reinforcing that while the Planck cosmology remains our adopted baseline for subsequent calculations, it is less consistent with these late-universe distance indicators.

Together, these results highlight the well-known *Hubble tension*: the persistent disagreement between the value of H_0 inferred from early-universe observations (such as the CMB) and that derived from late-universe distance measurements (e.g., supernovae or galaxy surveys). Despite this tension, we will adopt the fiducial Planck ΛCDM cosmology for the remainder of this work, while noting its inconsistency with the SNIa luminosity distances used in our MCMC analysis.

2. Recombination history

Having established the background cosmology, we now turn to the recombination history of the Universe. While the background evolution governs the large-scale expansion, it is the microphysics of the photon-baryon plasma that determines when and how the CMB photons we observe today were released. Three epochs are of particular importance: *recombination*, when free electrons and protons combine to form neutral hydrogen at $z \sim 1100$; *decoupling*, when the photon mean free path exceeds the Hubble horizon and the CMB photons stream freely for the first time; and *reionisation*, when the first generation of stars reionises the intergalactic medium at $z \sim 8$. Together, these epochs shape the optical depth of the Universe and leave distinct imprints on the CMB power spectrum. In this section we compute the free electron fraction X_e, the optical depth τ, and the visibility function $\tilde{g}$, which together encode the full scattering history of CMB photons. The theory and methodology follow Winther (2026a,c).

2.1. Theory

At early times, the Universe was so hot that it was fully ionised, with photons and baryons tightly coupled through Thomson scattering in a photon-baryon plasma. As the Universe expanded and cooled, free electrons and protons combined to form neutral hydrogen in a process known as *recombination*:

$$e^- + p^+ \rightleftharpoons H + \gamma.$$

Before recombination, the plasma was opaque to photons, which scattered repeatedly off free electrons and could not travel freely. After recombination, the photon mean free path grew to exceed the Hubble horizon, and the photons decoupled from the baryons, i.e. the Universe became transparent. The photons released at this moment of decoupling form the CMB we observe today, making the recombination history central to understanding the origin and properties of the CMB.

The opacity of a medium to radiation is quantified by the *optical depth* τ. A beam of light with initial intensity I_0 traversing a medium of optical depth τ emerges with intensity $I = I_0 e^{-\tau}$, so that $\tau \ll 1$ corresponds to a transparent medium and $\tau \gg 1$ to an opaque one. In cosmology, the dominant source of opacity is Thomson scattering of photons off free electrons, giving

$$\tau(\eta) = \int_{\eta}^{\eta_0} n_e \sigma_T a \, \mathrm{d}\eta',$$

where n_e is the free electron number density, σ_T is the Thomson cross-section[1], and a is the scale factor. The epoch of last scattering, from which the CMB photons we observe today last scattered, corresponds to $\tau \approx 1$.

In order to calculate the optical depth, we must therefore solve for the electron number density n_e, or in practice its proxy, the free electron fraction $X_e \equiv n_e/n_H$, where n_H corresponds to the number density of protons (hydrogen nuclei).

2.1.1. Electron fraction and number density

In the early Universe, when ionising radiation is still efficient and the plasma remains in thermal equilibrium, the free electron fraction can be determined from the *Saha equation*. As a first approximation, we neglect helium and assume all baryons are protons ($Y_p = 0$), so that the proton number density is

$$n_H \approx n_b = \frac{\rho_b}{m_H} = \frac{3H_0^2\Omega_{b0}}{8\pi G m_H a^3}.$$

Under this assumption, the Saha equation takes the form

$$\frac{X_e^2}{1-X_e} = \frac{1}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b}, \tag{13}$$

where m_e is the electron mass, k_B is Boltzmann's constant, $\epsilon_0 = 13.6$ eV is the ionisation energy of hydrogen, $\hbar$ is the reduced Planck constant, and T_b is the baryon temperature. In principle, T_b and the photon temperature T_γ should be evolved separately, but since Compton scattering keeps the two tightly coupled until well after recombination, it is an excellent approximation to set $T_b \simeq T_\gamma = T_{\mathrm{CMB},0}/a$ (Winther 2026c, see appendix).

Including helium ($Y_p \neq 0$) complicates the picture, as helium can be both singly and doubly ionised, contributing additional free electrons. We define the helium mass fraction as

$$Y_p \equiv \frac{m_{\mathrm{He}} n_{\mathrm{He}}}{m_b n_b} = \frac{4 n_{\mathrm{He}}}{n_b},$$

where we have approximated the baryon mass as $m_b \approx m_H$, the hydrogen atom mass, and used the fact that each helium nucleus contains four baryons. It follows that the hydrogen number density is

$$n_H = (1-Y_p)\, n_b = (1-Y_p)\frac{3H_0^2\Omega_{b0}}{8\pi G m_H a^3}.$$

To account for the three ionisation states of hydrogen and helium, we introduce the ionisation fractions

$$u \equiv \frac{n_{\mathrm{He}^+}}{n_{\mathrm{He}}}, \qquad v \equiv \frac{n_{\mathrm{He}^{2+}}}{n_{\mathrm{He}}}, \qquad w \equiv \frac{n_{\mathrm{H}^+}}{n_{\mathrm{H}}},$$

each of which satisfies its own set of three Saha equations

$$\frac{u}{1-u-v} = \frac{2}{n_e}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_0/k_B T_b}, \tag{14}$$

$$\frac{v}{u} = \frac{4}{n_e}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_1/k_B T_b}, \tag{15}$$

$$\frac{w}{1-w} = \frac{1}{n_e}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b}. \tag{16}$$

Here $\chi_0 = 24.587$ eV is the ionisation energy of neutral helium and $\chi_1 = 4\epsilon_0 = 54.418$ eV is the ionisation energy of He^+. By tallying the free electrons contributed by each ion species, we can define the quantity

$$f_e \equiv \frac{n_e}{n_b} = (2v+u)\,\frac{Y_p}{4} + w\,(1-Y_p), \tag{17}$$

from which the free electron fraction follows as $X_e = f_e/(1-Y_p)$. The factor of two in $2v$ reflects the fact that each doubly ionised helium nucleus contributes two free electrons.

The Saha approximation is valid as long as the plasma remains in thermal equilibrium, which holds well for $X_e \gtrsim 0.99$. As recombination proceeds and the Universe departs from equilibrium, the Saha equation overestimates the recombination rate and a non-equilibrium treatment is required. The *Peebles equation* (Peebles 1968) provides such a treatment, tracking the non-equilibrium evolution of X_e through the ODE

$$\frac{\mathrm{d}X_e}{\mathrm{d}x} = \frac{C_r(T_b)}{H}\left[\beta(T_b)(1-X_e) - n_H\alpha^{(2)}(T_b)X_e^2\right].$$

The two terms in brackets represent the competing processes of photoionisation (first term), which increases X_e, and recombination (second term), which decreases it. The prefactor C_r accounts for the fact that recombinations directly to the ground state produce an ionising photon that immediately re-ionises a nearby hydrogen atom, making such recombinations ineffective. Only recombinations to excited states, followed by either the two-photon $2s \to 1s$ decay or escape of a Lyman-α photon, result in a net increase of neutral hydrogen. The various quantities appearing in the Peebles equation are

$$\begin{aligned}
C_r(T_b) &= \frac{\Lambda_{2s\to1s} + \Lambda_\alpha}{\Lambda_{2s\to1s} + \Lambda_\alpha + \beta^{(2)}(T_b)},\\
\Lambda_{2s\to1s} &= 8.227\ \mathrm{s}^{-1},\\
\Lambda_\alpha &= \frac{H}{(8\pi)^2 n_{1s}}\left(\frac{3\epsilon_0}{\hbar c}\right)^3,\\
n_{1s} &= (1-X_e)n_H,\\
\beta^{(2)}(T_b) &= \beta(T_b)\, e^{3\epsilon_0/4k_B T_b},\\
\beta(T_b) &= \alpha^{(2)}(T_b)\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b},\\
\alpha^{(2)}(T_b) &= \frac{8\sigma_T c}{\sqrt{3\pi}}\sqrt{\frac{\epsilon_0}{k_B T_b}}\,\phi_2(T_b),\\
\phi_2(T_b) &= 0.448\ln(\epsilon_0/k_B T_b)\,.
\end{aligned}$$

Here $\Lambda_{2s\to1s}$ is the decay rate of the $2s$ state via two-photon emission, Λ_α is the Lyman-α escape rate, $\alpha^{(2)}$ is the recombination coefficient to excited states ($n \geq 2$), β is the corresponding photoionisation rate, and $\beta^{(2)}$ is the photoionisation rate from the $2s$ state.

2.1.2. Reionisation

Around a redshift of $z \sim 10$, the first stars and galaxies form and begin emitting large amounts of high-energy ultraviolet radiation. This radiation is energetic enough to ionise the surrounding neutral hydrogen and helium, driving the free electron fraction X_e back up from its post-recombination value of $X_e \sim 10^{-4}$ toward values close to unity. This process is known as *reionisation*. Our primary interest is its effect on CMB photons propagating

[1] The Thomson scattering cross-section is given by $\sigma_T = \dfrac{8\pi}{3}\dfrac{\alpha^2\hbar^2}{m_e^2c^2}$.

from the surface of last scattering to us today. For this purpose a simple phenomenological model suffices. We will thus use the same simplified model used in CAMB Lewis et al. (2000), adding two tanh-shaped transitions to the Peebles solution

$$X_e = X_e^{\mathrm{Peebles}} + \frac{1+f_{\mathrm{He}}}{2}\left(1+\tanh\frac{y_{\mathrm{reion}}-y}{\Delta y_{\mathrm{reion}}}\right) + \frac{f_{\mathrm{He}}}{2}\left(1+\tanh\frac{z_{\mathrm{Hereion}}-z}{\Delta z_{\mathrm{Hereion}}}\right). \quad (18)$$

The first term accounts for the reionisation of hydrogen and the second for the reionisation of helium. Each transition is centred on its respective reionisation redshift and smoothed over a width Δz. Here $y \equiv (1+z)^{3/2} = e^{-3x/2}$ is a rescaled redshift variable, $f_{\mathrm{He}} \equiv Y_p/[4(1-Y_p)]$ is the helium-to-hydrogen number ratio, $y_{\mathrm{reion}} = (1+z_{\mathrm{reion}})^{3/2}$, and $\Delta y_{\mathrm{reion}} = \frac{3}{2}\sqrt{1+z_{\mathrm{reion}}}\,\Delta z_{\mathrm{reion}}$ is the corresponding width in y-space. From the assumed Planck cosmology, we have $z_{\mathrm{reion}} = 8$, $\Delta z_{\mathrm{reion}} = 0.5$, $z_{\mathrm{Hereion}} = 3.5$, and $\Delta z_{\mathrm{Hereion}} = 0.5$.

2.1.3. Visibility function

After we have solved for the optical depth τ, we will also compute the *visibility function* $\tilde{g}$ given by

$$\tilde{g}(x) = -\tau'(x)e^{-\tau(x)}. \quad (19)$$

It describes the probability density for a given photon to last time having scattered at time x, and has the property $\int_{-\infty}^{0}\tilde{g}(x)\mathrm{d}x = 1$. The visibility function peaks when the photons decoupled from the baryons, i.e. at the transition between the opaque and transparent epochs (taken to be when $\tau = 1$).

2.1.4. Sound horizon

The final quantity of interest is the *sound horizon* at decoupling which is the total comoving distance a sound wave in the photon-baryon plasma of the early Universe could travel from the Big Bang until photons decouple. This scale length is the maximum distance between regions that could have had causal contact and will thus leave a distinct imprint on both the CMB power spectrum and the large-scale distribution of matter in the Universe. The sound speed of the tightly coupled photon-baryon plasma is somewhat lower than the relativistic sound speed of a pure photon gas ($c/\sqrt{3}$), and is given by

$$c_s = \frac{c\sqrt{R}}{\sqrt{3(1+R)}}, \qquad R \equiv \frac{4\Omega_{\gamma 0}}{3\Omega_{b0}a}.$$

The sound horizon is then

$$s(x) = \int_{-\infty}^{x}\frac{c_s}{\mathcal{H}}\,\mathrm{d}x', \quad \Rightarrow \quad \frac{\mathrm{d}s}{\mathrm{d}x} = \frac{c_s}{\mathcal{H}(x)}, \quad (20)$$

and its value at decoupling $s(x_{\mathrm{dec}})$ sets the physical scale of the acoustic peaks in the CMB.

2.2. Implementation details

2.2.1. Solving for the free electron fraction

As outlined in Section 2.1.1, the evolution of the free electron fraction X_e is divided into three distinct epochs, each requiring a different treatment; the equilibrium Saha regime at early times, the out-of-equilibrium Peebles regime during and after recombination, and the reionisation epoch at late times. The calculation of X_e is therefore done in three phases:

(i) first we calculate X_e in Saha regime, using either Eq.(13) or Eqs.(14) to (16) depending on if we include helium or not, over the whole time interval, spanning from $x_{\mathrm{start}} = \ln(10^{-8})$ up to $x_{\mathrm{end}} = 0$;

(ii) then we search through the Saha solution and identify when the approximation breaks down (i.e. when $X_e < 0.99$) and start solving the Peebles ODE, Eq.(2.1.1), from this point onwards, replacing the values from the Saha solution along the way;

(iii) lastly, we the reionisation contribution at late times to the Peebles solution, as described by Eq.(18), and spline the result at the end.

The hydrogen-only Saha equation, Eq. (13), reduces to a simple quadratic in X_e, which can be solved analytically. Retaining the physically meaningful positive root, we obtain

$$X_e = \frac{-A_s + A_s\sqrt{1+4/A_s}}{2},$$

where $A_s(T_b)$ is just the RHS of Eq.(13). Going far back in time, the baryon temperature will get extremely large such that $A_s \sim T_b^{3/2}$ blows up and may cause overflows. To get around this problem we Taylor expand the square root term[2] and find that $X_e = 1$ in this limit. This is an excellent approximation as long as $A_s \gg 4$. In our numerical implementation, the approximate solution is imposed when $A_s > 400$.

If however we include helium in our Universe ($Y_p \neq 0$), we must instead solve the three Saha equations, Eqs.(14) to (16). Defining the quantities

$$\begin{aligned} r_0 &\equiv \frac{2}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_0/k_B T_b}, \\ r_1 &\equiv \frac{4}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\chi_1/k_B T_b}, \\ r_2 &\equiv \frac{1}{n_b}\left(\frac{m_e k_B T_b}{2\pi\hbar^2}\right)^{3/2} e^{-\epsilon_0/k_B T_b}, \end{aligned}$$

we can solve for the ion matter fractions u, v, and w as given by

$$\begin{aligned} u &= \frac{f_e r_0}{f_e^2 + f_e r_0 + r_0 r_1}, \\ v &= \frac{r_0 r_1}{f_e^2 + f_e r_0 + r_0 r_1}, \\ w &= \frac{r_2}{f_e + r_2}. \end{aligned}$$

Numerically, the system is solved iteratively at each time step. Beginning with an initial guess of $f_e = 1$, we solve for the ionisation fractions u, v, and w from the three Saha equations, then update f_e using Eq. (17). This cycle is repeated until convergence, defined by $|f_{e,\mathrm{new}} - f_{e,\mathrm{old}}| < 10^{-10}$, which is typically achieved within a handful of iterations. The free electron fraction then follows as $X_e = f_e/(1-Y_p)$.

Going into the Peebles regime, we set the initial condition of Eq.(2.1.1) to be the value of X_e from the last valid Saha solution

[2] $\sqrt{1+x} \simeq 1 + x/2$ when $|x| \ll 1$.

to ensure continuity in the full solution. The only thing worth noting about the numerical implementation is that $\beta^{(2)}$ is set to zero when $\epsilon_0/k_B T_b \gg 1$ to avoid numerical overflow from the exponential factor. Specifically, we set $\beta^{(2)} = 0$ when $\epsilon_0/k_B T_b > 200$.

Finally, introducing the reionisation contributions from hydrogen and helium reionisation is as simple as adding the extra tanh-terms in Eq.(18). After that is done, we spline the resulting function X_e and n_e in log-space using cubic spline, since their values span several orders of magnitude within short time frames.

2.2.2. Calculating the optical depth and visibility function

Once we have solved for the free electron fraction, and have thus also by proxy calculated the free electron number density n_e, we calculate the optical depth from its differential form

$$\tau' = \frac{\mathrm{d}\tau}{\mathrm{d}x} = -\frac{c\sigma_T n_e(x)}{H(x)},$$

imposing the intial condition $\tau(0) = 0$. This is achieved by solving the ODE backwards in time from $x_{\mathrm{end}} = 0$ to x_{start}. We spline the result and its derivative, and calculate its second derivative by differentiating the τ' using the method built into our spline routine (Winther 2026g).

The visibility function is then calculated using Eq.(19). Splining $\tilde{g}$, we then its first two derivatives, $\tilde{g}'$ and $\tilde{g}''$, employing the same differentiating scheme for cubic splines.

2.2.3. Calculating the sound horizon at decoupling

To get the sound horizon as a function of x, we solve the ODE stated in Eq.(20) using the usual method. The initial condition is found deep in the radiation dominated era of the Universe, when $\mathcal{H}(x) \sim H_0\sqrt{\Omega_{R0}}e^{-x}$ and $R \to \infty$. The sound speed of the photon-baryon plasma is slowly varying, and approximately constant, since

$$\lim_{x\to-\infty} c_s(x) = \frac{c}{\sqrt{3}} \lim_{R\to\infty} \frac{1}{\sqrt{1+1/R}} = \frac{c}{\sqrt{3}}.$$

We then get the initial condition

$$s(x_{start}) = \frac{1}{H_0\sqrt{\Omega_{R0}}} \int_{-\infty}^{x_{\mathrm{start}}} c_s(x)\, e^x \mathrm{d}x = \frac{c_s(x_{\mathrm{start}})}{\mathcal{H}(x_{\mathrm{start}})}.$$

After splining the result, we plug in the value of x at the time of decoupling, defined as when $\tau(x_{\mathrm{dec}}) = 1$ or $\tilde{g}$ is maximal, to get the sound horizon at decoupling $r_s \equiv s(x_{\mathrm{dec}})$.

2.3. Results

2.3.1. The free electron fraction and recombination

Figure 9 displays the evolution of the free electron fraction X_e as a function of x in three different situations; the first is the fiducial model which takes into account both hydrogen and helium ionisation (with $Y_p = 0.245$) and reionisation, the second is the simplest model, corresponding to a hydrogen only Universe with no reionisation, and the third is the pure Saha solution for a hydrogen and helium Universe ($Y_p = 0.245$) with no reionisation.

The top panel shows the overall evolution of X_e. At early times the plasma is fully ionised, with helium doubly ionised ($u = 0,\ v = 1,\ w = 1$), giving

$$X_e = \frac{f_e}{1-Y_p} = \frac{1-Y_p/2}{1-Y_p} \simeq 1.162.$$

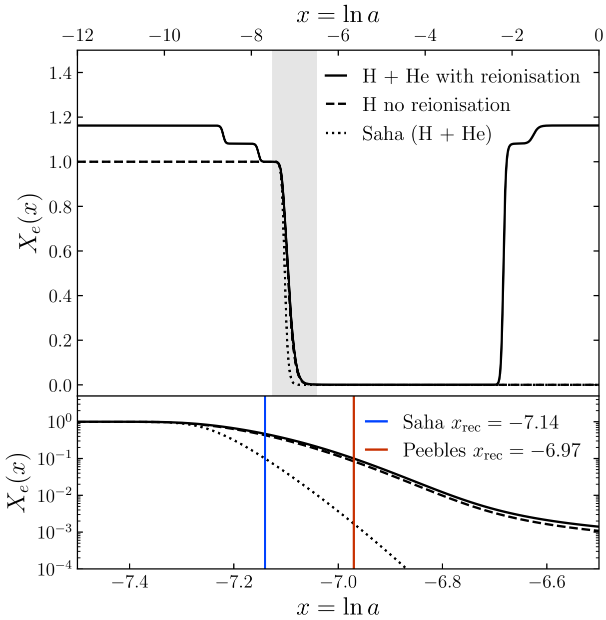

Fig. 9. The free electron fraction X_e as a function of $x = \ln a$ for three configurations: hydrogen and helium with reionisation (solid), hydrogen only without reionisation (dashed), and the pure Saha approximation with both species and without reionisation (dotted), with $Y_p = 0.245$ where applicable. The bottom panel magnifies the recombination epoch ($-7.5 \leq x \leq -6.5$, shaded grey in the top panel), highlighting the different recombination times.

As the Universe cools and $k_B T_b \lesssim \chi_1$, helium transitions to being singly ionised ($u = 1,\ v = 0,\ w = 1$), reducing $X_e \simeq 1.081$ at around $x \sim -8.7$. A further drop occurs when $k_B T_b \lesssim \chi_0$, at which point helium becomes fully neutral and only hydrogen remains ionised ($X_e = 1$). The final and most dramatic decrease corresponds to hydrogen recombination, after which X_e freezes out at a relic plateau of $\sim 10^{-4}$. In the reionisation model, X_e rises sharply back toward the initial $X_e \simeq 1.162$ at $x \sim -2.5$ as the first stars reionise the intergalactic medium.

The hydrogen-only model without reionisation follows the same qualitative behaviour, but lacks both the early helium transitions and the late-time reionisation upturn.

The bottom panel of Figure 9 magnifies the hydrogen recombination epoch ($-7.5 \leq x \leq -6.5$) on a logarithmic scale, allowing a clearer comparison between the Saha and Peebles solutions. The pure Saha approximation predicts a significantly faster decline in X_e, continuing to decrease toward machine zero as it assumes thermal equilibrium is maintained throughout. This leads to an earlier predicted time of recombination (defined as when $X_e = 0.1$) with $x_{\mathrm{rec}} = -7.14$ compared to $x_{\mathrm{rec}} = -6.97$ from the Peebles solution. The two predictions are compared in terms of x, redshift z, and cosmic time t in Table 2.

The final freeze-out abundance of free electrons today was calculated to be $X_e(0) = 2.674 \times 10^{-4}$ in the fiducial model ($Y_p = 0.245$) without reionisation.

2.3.2. The optical depth and visibility function

Figure 10 shows the optical depth and its first two derivatives with respect to x for four configurations: with and without helium ($Y_p = 0.245$ or $Y_p = 0$), and with and without reionisation.

Decoupling and recombination times ($Y_p = 0.245$)				
Event		Log scale factor x	Redshift z	Time t [Myr]
Decoupling	Peebles	-6.99474	1089.88	0.37211
	Saha	-7.17033	1299.28	0.276427
Recombination	Peebles	-6.97013	1063.36	0.3878
	Saha	-7.1404	1260.87	0.2909

Table 2. The times for decoupling and recombination for the pure Saha solution and the Peebles solution expressed in terms of x, redshift z, and cosmic time t.

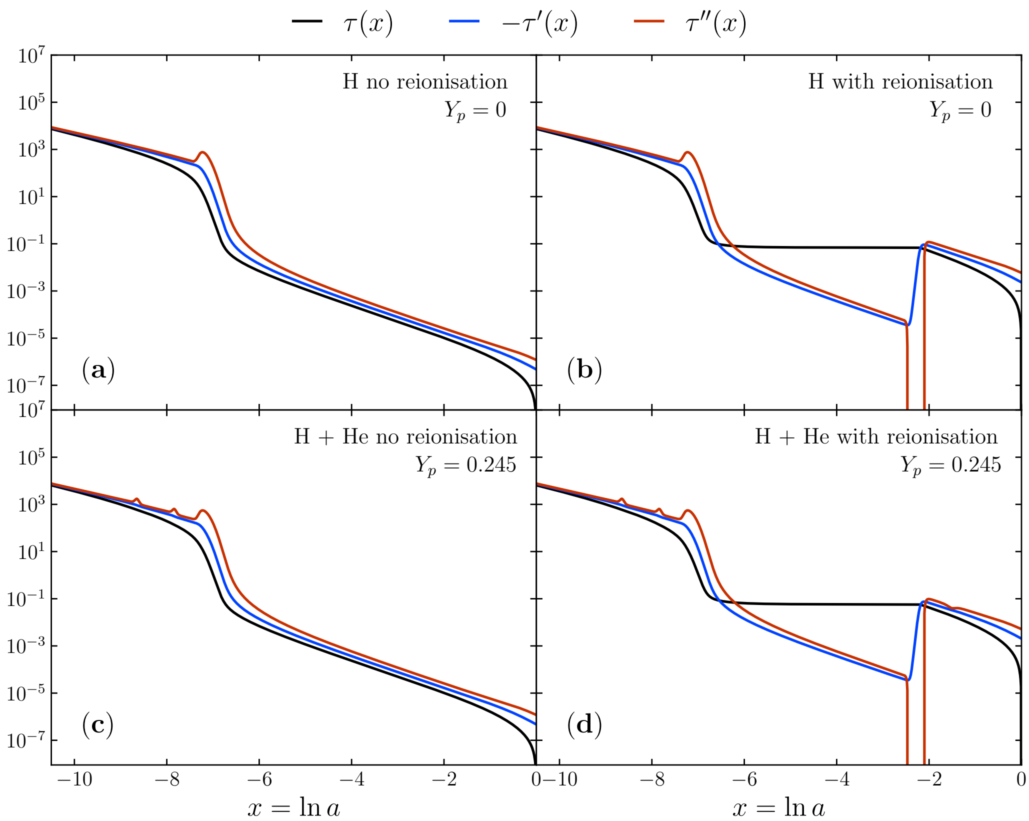

Fig. 10. Optical depth τ and its derivatives τ', τ'' as function of x for four different configurations; either with or without helium, and with or without reionisation at late times.

Panel (a), the hydrogen-only case without reionisation, shows a smooth monotonically increasing τ going back in time, crossing $\tau = 1$ near the epoch of recombination, beyond which the Universe is optically thick ($\tau \gg 1$).

Introducing reionisation, as in panels (b) and (d), imprints a secondary feature at $x \sim -2.5$ where the sudden rise in X_e causes τ to plateau at $\tau \sim 0.1$ between the reionisation epoch and decoupling, reflecting the additional Thomson scattering experienced by CMB photons as they traverse the reionised intergalactic medium.

Including helium, as in panels (c) and (d), leaves its most visible imprint on the second derivative τ'', which develops additional small bumps at the epochs of helium recombination and reionisation. These correspond directly to the stepwise drops seen in X_e in Figure 9, where each transition in the helium ionisation state produces a brief change in the rate at which τ evolves.

Figure 11 shows the visibility function $\tilde{g}$ and its first two derivatives for the fiducial model with both hydrogen and helium. The visibility function is sharply peaked at $x_{\rm dec} = -6.973$, marking the epoch of decoupling, commonly referred to as the surface of last scattering, where the probability of a CMB photon undergoing its last Thomson scattering is maximised. The sharpness of this peak reflects the relatively brief duration of recombination: although recombination is not instantaneous, the transition from an opaque to a transparent Universe occurs over a narrow range of x, so the surface of last scattering is well-defined.

The small bump visible at $x \sim -2.5$ corresponds to reionisation, where the rise in X_e briefly increases the scattering rate and produces a small but non-zero probability of late-time scat-

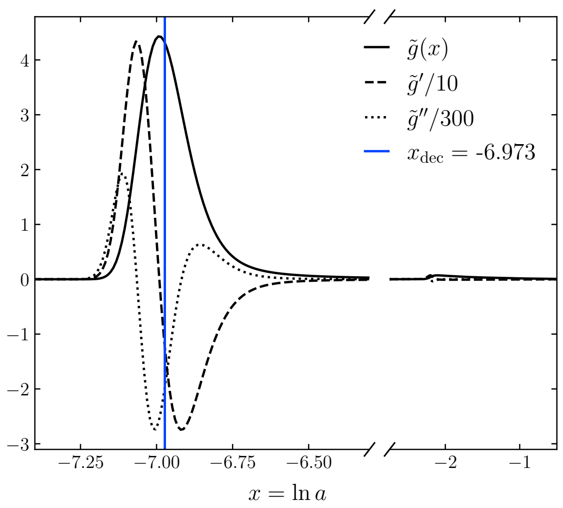

Fig. 11. Visibility function $\tilde{g}$ and its two first derivatives $\tilde{g}'$, $\tilde{g}''$ as functions of x for the Universe with both hydrogen and helium ($Y_p = 0.245$) and late-stage reionisation.

tering. Since $\tilde{g}$ represents the probability density that a photon was last scattered at a given x, it must satisfy the normalisation condition $\int_{-\infty}^{0} \tilde{g}(x)\,\mathrm{d}x = 1$, which our numerical implementation reproduces to machine precision, confirming the consistency of our solution. The derivatives $\tilde{g}'$ and $\tilde{g}''$ are smooth and well-behaved, and will be needed as source terms when computing CMB temperature and polarisation anisotropies. The decoupling times predicted by the Saha approximation and the Peebles equation are compared in Table 2. The full Peebles solution better matches the ΛCDM prediction of $z_{\mathrm{rec}} \sim 1100$.

2.3.3. Sound horizon at decoupling

Evaluating the sound horizon at decoupling r_s for the fiducial model, i.e. at $x_{\mathrm{dec}} = -6.99474$ (corresponding to redshift $z_{\mathrm{dec}} = 1089.88$), we find

$$r_s \equiv s(x_{\mathrm{dec}}) \simeq 144.413 \text{ Mpc}. \tag{21}$$

This value is in good agreement with the standard result from *Planck* (Aghanim et al. 2020), and sets a fundamental physical scale in the Universe. At decoupling there is no longer any radiative pressure to counteract the gravitational collapse of the baryons, so the acoustic oscillations in the photon-baryon plasma are frozen in place, and the sound horizon marks the maximum distance over which these oscillations could have coherently developed. This scale is directly imprinted as the characteristic angular separation of the acoustic peaks in the CMB power spectrum, and as the baryon acoustic oscillation (BAO) scale in the matter power spectrum.

3. Perturbations

In this section we perturb the metric and the distribution functions of each species around their homogeneous backgrounds and study how these perturbations evolve. We first introduce the theoretical framework, covering the perturbed metric in Newtonian gauge, the Boltzmann hierarchies governing each species, and the adiabatic initial conditions from inflation. We then describe the numerical implementation, including the tight-coupling approximation used to stabilise the integration through recombination, before presenting the resulting evolution of the matter, metric, and polarisation perturbations.

3.1. Theory

The perturbation theory presented in this section is heavily inspired by Callin (2006) and Winther (2026d,a). We will mainly cover the notation used as well as the main equations governing the pertubations without diving too much into the details. For a more thorough derivation of all the perturbation equations, it is therefore suggested to read the lecture notes by Winther (2026a).

3.1.1. Setting up the perturbations

Working in Newtonian gauge, we perturb the FLRW metric, Eq.(1), by introducing two scalar gravitational potentials $\Psi(t, \boldsymbol{x})$ and $\Phi(t, \boldsymbol{x})$, giving the line element

$$\mathrm{d}s^2 = -(1 + 2\Psi)\,\mathrm{d}t^2 + a^2(1 + 2\Phi)\left(\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2\right). \tag{22}$$

Both potentials are closely related to the Newtonian gravitational potential. We restrict ourselves to scalar perturbations, which dominate our Universe and are therefore the most cosmologically relevant for our model.

Perturbations to the photon distribution are described in Fourier space by the relative temperature fluctuation

$$\Theta(t, \boldsymbol{k}, \mu) \equiv \frac{\delta T_{\mathrm{CMB}}(t, \boldsymbol{k}, \mu)}{\bar{T}_{\mathrm{CMB}}(t)},$$

so that $T_{\mathrm{CMB}} = \bar{T}_{\mathrm{CMB}}(1 + \Theta)$. Here $\boldsymbol{k}$ is the Fourier-space conjugate to the spatial position vector $\boldsymbol{x}$, and $\mu \equiv \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{p}}$ is the cosine of the angle between $\boldsymbol{k}$ and the photon momentum $\boldsymbol{p}$. We work in Fourier space because the spatial homogeneity of the background ensures that different Fourier modes $\boldsymbol{k}$ evolve independently, turning the perturbation PDEs into a separate ODE system for each k.

Crucially, Θ depends only on the direction of $\boldsymbol{p}$, not its magnitude; it is this directional dependence that gives rise to the temperature anisotropies observed in the CMB. This motivates expanding the photon perturbation in multipoles

$$\Theta_\ell(t, k) = \frac{i^\ell}{2} \int_{-1}^{1} P_\ell(\mu)\Theta(t, k, \mu)\mathrm{d}\mu,$$

using the Legendre polynomials $P_\ell(\mu)$ (see Appendix C). The expansion of Θ in the multipoles is then given by

$$\Theta(t, k, \mu) = \sum_{\ell=0}^{\infty} \frac{2\ell + 1}{i^\ell} \Theta_\ell(t, k) P_\ell(\mu). \tag{23}$$

The same multipole expansion will be done for the perturbations to the photon polarisation Θ_P and the neutrino perturbations $\mathcal{N}$.

3.1.2. The equations of motion for the perturbations

For each species in our model (CDM, baryons, photons, neutrinos) the phase-space evolution is governed by a Boltzmann equation of the form

$$\frac{\mathrm{d}f_i}{\mathrm{d}t} = C_i,$$

where f_i is the distribution function of species i[3] and C_i is the collision term encoding all relevant interactions. In a cosmological context, the dominant processes are Compton scattering, $e^- + \gamma \rightleftharpoons e^- + \gamma$, and Coulomb scattering, $e^- + p^+ \rightleftharpoons e^- + p^+$, which tightly couple the photon, electron, and proton fluids at early times. Perturbing each distribution function $f_i \to \bar{f}_i + \delta f_i$ and evaluating the Boltzmann equation then yields the temporal evolution of each species.

The metric perturbations in Eq. (22) enter the Einstein field equations, Eq. (2), from both sides simultaneously. On the left-hand side, the perturbed metric modifies the Christoffel symbols, which propagate through to perturb the Ricci tensor and ultimately the Einstein tensor, $G_{\mu\nu} \to \bar{G}_{\mu\nu} + \delta G_{\mu\nu}$. On the right-hand side, the perturbed distribution functions yield a perturbed stress-energy tensor, $T_{\mu\nu} \to \bar{T}_{\mu\nu} + \delta T_{\mu\nu}$. Matching at first order in the perturbations then gives a closed set of equations governing the evolution of the gravitational potentials Ψ and Φ.

For CDM and baryons, the first and second moments of the perturbation equations describe the evolution of the density contrasts $\delta_{\rm CDM}$, δ_b and the bulk velocities $v_{\rm CDM}$, v_b respectively. CDM is collisionless and only couples gravitationally, while baryons receive an additional momentum boost from Compton scattering off photons, encoded in the $\tau' R(3\Theta_1 + v_b)$ term,

$$
\begin{aligned}
\delta'_{\rm CDM} &= \frac{ck}{\mathcal{H}} v_{\rm CDM} - 3\Phi', \\
v'_{\rm CDM} &= -v_{\rm CDM} - \frac{ck}{\mathcal{H}}\Psi, \\
\delta'_b &= \frac{ck}{\mathcal{H}} v_b - 3\Phi', \\
v'_b &= -v_b - \frac{ck}{\mathcal{H}}\Psi + \tau' R(3\Theta_1 + v_b).
\end{aligned}
$$

Here $R = 4\Omega_{\gamma 0}/3\Omega_{b0}a$, and the apostrophes refer to derivatives with respect to $x = \ln a$.

The photon temperature perturbations are described by an infinite Boltzmann hierarchy in the multipole moments Θ_ℓ, where $\ell = 0, 1, 2$ correspond to the monopole (mean temperature), dipole (bulk flow) and quadrupole (anisotropic stress), respectively. Compton scattering damps higher multipoles and couples the photon dipole to the baryon velocity v_b. The hierarchy is truncated at $\ell = \ell_{\rm max}$ using a closure relation that minimises reflections,

$$
\begin{aligned}
\Theta'_0 &= -\frac{ck}{\mathcal{H}}\Theta_1 - \Phi', \\
\Theta'_1 &= \frac{ck}{3\mathcal{H}}\left[\Theta_0 - 2\Theta_2 + \Psi\right] + \tau'\left[\Theta_1 + \frac{1}{3}v_b\right], \\
\Theta'_\ell &= \frac{ck}{(2\ell+1)\mathcal{H}}\left[\ell\Theta_{\ell-1} - (\ell+1)\Theta_{\ell+1}\right] + \tau'\left[\Theta_\ell - \frac{1}{10}\Pi\,\delta_{\ell,2}\right] \\
&\quad \text{for } 2 \le \ell < \ell_{\rm max}, \\
\Theta'_\ell &= \frac{ck}{\mathcal{H}}\left[\Theta_{\ell-1} - \frac{\ell+1}{k\eta(x)}\Theta_\ell\right] + \tau'\Theta_\ell \quad \text{for } \ell = \ell_{\rm max},
\end{aligned}
$$

where $\Pi \equiv \Theta_2 + \Theta_{P0} + \Theta_{P2}$ is the polarisation source term. The photon polarisation perturbations $\Theta_{P\ell}$ obey a similar hierarchy, sourced by Π through Compton scattering,

$$
\begin{aligned}
\Theta'_{P0} &= -\frac{ck}{\mathcal{H}}\Theta_{P1} + \tau'\left[\Theta_{P0} - \frac{1}{2}\Pi\right], \\
\Theta'_{P\ell} &= \frac{ck}{(2\ell+1)\mathcal{H}}\left[\ell\Theta_{P,\ell-1} - (\ell+1)\Theta_{P,\ell+1}\right] + \tau'\left[\Theta_{P\ell} - \frac{1}{10}\Pi\,\delta_{\ell,2}\right] \\
&\quad \text{for } 1 \le \ell < \ell_{\rm max}, \\
\Theta'_{P,\ell} &= \frac{ck}{\mathcal{H}}\left[\Theta_{P,\ell-1} - \frac{\ell+1}{k\eta(x)}\Theta_{P\ell}\right] + \tau'\Theta_{P\ell} \quad \text{for } \ell = \ell_{\rm max}.
\end{aligned}
$$

Since neutrinos are collisionless after decoupling, their Boltzmann hierarchy takes the same form as the photon hierarchy but with no collision terms,

$$
\begin{aligned}
\mathcal{N}'_0 &= -\frac{ck}{\mathcal{H}}\mathcal{N}_1 - \Phi', \\
\mathcal{N}'_1 &= \frac{ck}{3\mathcal{H}}\left[\mathcal{N}_0 - 2\mathcal{N}_2 + \Psi\right], \\
\mathcal{N}'_\ell &= \frac{ck}{(2\ell+1)\mathcal{H}}\left[\ell\mathcal{N}_{\ell-1} - (\ell+1)\mathcal{N}_{\ell+1}\right] \quad \text{for } 2 \le \ell < \ell_{\rm max,\nu}, \\
\mathcal{N}'_\ell &= \frac{ck}{\mathcal{H}}\left[\mathcal{N}_{\ell-1} - \frac{\ell+1}{k\eta(x)}\mathcal{N}_\ell\right] \quad \text{for } \ell = \ell_{\rm max,\nu}.
\end{aligned}
$$

Finally, the gravitational potentials are sourced by the total energy density and momentum of all species through the Einstein field equations. The equation for Φ' is a constraint relating the time derivative of the potential to the density perturbations of all components, while Ψ differs from $-\Phi$ by a term proportional to the anisotropic stress sourced by the photon and neutrino quadrupoles Θ_2 and $\mathcal{N}_2$,

$$
\begin{aligned}
\Phi' &= \Psi - \frac{1}{3}\left(\frac{ck}{\mathcal{H}}\right)^2\Phi \\
&\quad + \frac{H_0^2}{2\mathcal{H}^2 a^2}\left[\Omega_{\rm CDM0}a\delta_{\rm CDM} + \Omega_{b0}a\delta_b + 4\Omega_{\gamma 0}\Theta_0 + 4\Omega_{\nu 0}\mathcal{N}_0\right], \\
\Psi &= -\Phi - \frac{12H_0^2}{c^2k^2a^2}\left[\Omega_{\gamma 0}\Theta_2 + \Omega_{\nu 0}\mathcal{N}_2\right].
\end{aligned} \tag{24}
$$

3.1.3. Initial conditions from inflation

The system is initialized sufficiently deep in the radiation-dominated era such that we can assume that all modes of interest are superhorizon ($ck \ll \mathcal{H}$). The adiabatic initial conditions derived from inflationary models are

$$
\begin{aligned}
\Psi &= -\frac{1}{\frac{3}{2} + \frac{2f_\nu}{5}}, \\
\Phi &= -\left(1 + \frac{2f_\nu}{5}\right)\Psi, \\
\delta_{\rm CDM} = \delta_b &= -\frac{3}{2}\Psi, \\
v_{\rm CDM} = v_b &= -\frac{ck}{2\mathcal{H}}\Psi,
\end{aligned}
$$

where $f_\nu = \Omega_{\nu 0}/\Omega_{R0}$.

[3] The distribution function takes the Fermi-Dirac form $f = \left[\exp\left(\frac{E-\mu}{k_BT}\right) + 1\right]^{-1}$ for fermionic species, and the Bose-Einstein form $f = \left[\exp\left(\frac{E-\mu}{k_BT}\right) - 1\right]^{-1}$ for bosonic species.

For the photon multipoles,

$$\Theta_0 = -\frac{1}{2}\Psi,$$
$$\Theta_1 = \frac{ck}{6\mathcal{H}}\Psi,$$
$$\Theta_2 = \begin{cases} -\dfrac{8ck}{15\mathcal{H}\tau'}\,\Theta_1 & \text{(with polarisation)},\\ -\dfrac{20ck}{45\mathcal{H}\tau'}\,\Theta_1 & \text{(without polarisation)},\end{cases}$$
$$\Theta_\ell = -\frac{\ell}{2\ell+1}\frac{ck}{\mathcal{H}\tau'}\,\Theta_{\ell-1}, \qquad \ell \geq 3.$$

For the photon polarisation multipoles,

$$\Theta_0^P = \frac{5}{4}\,\Theta_2,$$
$$\Theta_1^P = -\frac{ck}{4\mathcal{H}\tau'}\,\Theta_2,$$
$$\Theta_2^P = \frac{1}{4}\,\Theta_2,$$
$$\Theta_\ell^P = -\frac{\ell}{2\ell+1}\frac{ck}{\mathcal{H}\tau'}\,\Theta_{\ell-1}^P, \qquad \ell \geq 3.$$

For the neutrino multipoles,

$$\mathcal{N}_0 = -\frac{1}{2}\Psi,$$
$$\mathcal{N}_1 = \frac{ck}{6\mathcal{H}}\Psi,$$
$$\mathcal{N}_2 = -\frac{c^2k^2a^2(\Phi+\Psi)}{12H_0^2\,\Omega_{\nu 0}},$$
$$\mathcal{N}_\ell = \frac{ck}{(2\ell+1)\mathcal{H}}\,\mathcal{N}_{\ell-1}, \qquad \ell \geq 3.$$

3.2. Implementation details

3.2.1. Tight coupling regime

Before photon decoupling, Thomson scattering couples the photon and baryon fluids so strongly that integrating the full ODE system becomes stiff and numerically unstable. We therefore adopt a *tight-coupling approximation*, in which the Boltzmann hierarchy for photons is truncated to only the monopole Θ_0 and dipole Θ_1 as independent degrees of freedom. All higher multipoles are depend on these via

$$\Theta_\ell \approx -\frac{\ell}{2\ell+1}\frac{ck}{\mathcal{H}\tau'}\Theta_{\ell-1}, \qquad \ell \geq 2,$$

and photon polarisation is neglected entirely. The evolution equations for Θ_1 and v_b are also modified: rather than integrating the full expressions, one instead solves a combined photon–baryon momentum equation in which the tight-coupling limit is taken analytically before being passed to the integrator (Winther 2026d),

$$q = \frac{-[(1-R)\tau' + (1+R)\tau''](3\Theta_1 + v_b)}{(1+R)\tau' + \frac{\mathcal{H}'}{\mathcal{H}} - 1} + \frac{-\frac{ck}{\mathcal{H}}\Psi + (1-\frac{\mathcal{H}'}{\mathcal{H}})\frac{ck}{\mathcal{H}}(-\Theta_0 + 2\Theta_2) - \frac{ck}{\mathcal{H}}\Theta_0'}{(1+R)\tau' + \frac{\mathcal{H}'}{\mathcal{H}} - 1},$$

such that

$$v_b' = \frac{1}{1+R}\left[-v_b - \frac{ck}{\mathcal{H}}\Psi + R(q + \frac{ck}{\mathcal{H}}(-\Theta_0 + 2\Theta_2) - \frac{ck}{\mathcal{H}}\Psi)\right],$$
$$\Theta_1' = \frac{1}{3}(q - v_b').$$

The expressions for the higher-ordered photon moments are the same in the tight-coupling regime as given by the initial conditions

$$\Theta_2 = \begin{cases} -\frac{8ck}{15\mathcal{H}\tau'}\Theta_1, & \text{(with polarisation)}\\ -\frac{20ck}{45\mathcal{H}\tau'}\Theta_1, & \text{(without polarisation)}\end{cases}$$
$$\Theta_\ell = -\frac{\ell}{2\ell+1}\frac{ck}{\mathcal{H}\tau'}\Theta_{\ell-1}, \qquad \ell > 2$$
$$\Theta_0^P = \frac{5}{4}\Theta_2$$
$$\Theta_1^P = -\frac{ck}{4\mathcal{H}\tau'}\Theta_2$$
$$\Theta_2^P = \frac{1}{4}\Theta_2$$
$$\Theta_\ell^P = -\frac{\ell}{2\ell+1}\frac{ck}{\mathcal{H}\tau'}\Theta_{\ell-1}^P, \qquad \ell > 2.$$

We impose two criteria for the tight-coupling regime to hold. First, the scattering rate must be large compared to the expansion rate and the oscillation wavenumber,

$$\left|\frac{\mathrm{d}\tau}{\mathrm{d}x}\right| > 10\min\left(1, \frac{ck}{\mathcal{H}}\right).$$

Second, we must be prior to the onset of recombination, which we take to occur at $x = -8.3$, so the approximation is applied only for $x < -8.3$.

3.2.2. Boltzmann hierarchy cutoff

As discussed in Section 3.1.2, we truncate the photon, photon polarisation, and neutrino hierarchies at $\ell_{\max}$ and $\ell_{\max,\nu}$ respectively. Although the CMB power spectrum eventually requires multipoles up to $\ell \sim 2000$, these high-ℓ moments need not be obtained from the ODE system directly: they are instead recovered via the *line-of-sight* integral (discussed in Section 4.1.2), which requires only Π, Φ, Ψ, Θ_0, and v_b. The highest multipole that must therefore be resolved accurately by the ODE solver is Θ_2, which enters the source term through Π. We therefore set $\ell_{\max} = \ell_{\max,\nu} = 8$, which is sufficient to ensure that all quantities entering the LOS integral are computed to adequate accuracy.

3.3. Results

3.3.1. Matter perturbations and the Baryon Acoustic Oscillations

Figure 12 shows the matter perturbations as functions of the log-scale factor x and wavenumber k. A feature common to all panels is that perturbations remain frozen at their inflationary initial values until the mode enters the horizon at $k \sim \mathcal{H}/c$: superhorizon modes ($k \ll \mathcal{H}/c$) are causally disconnected and cannot evolve, whereas subhorizon modes ($k \gg \mathcal{H}/c$) are free to collapse and oscillate.

Panel (a) displays the photon density contrast $\delta_\gamma = 4\Theta_0$ along with the neutrino denisty contrast $\delta_\nu = 4\mathcal{N}_0$. Upon horizon entry,

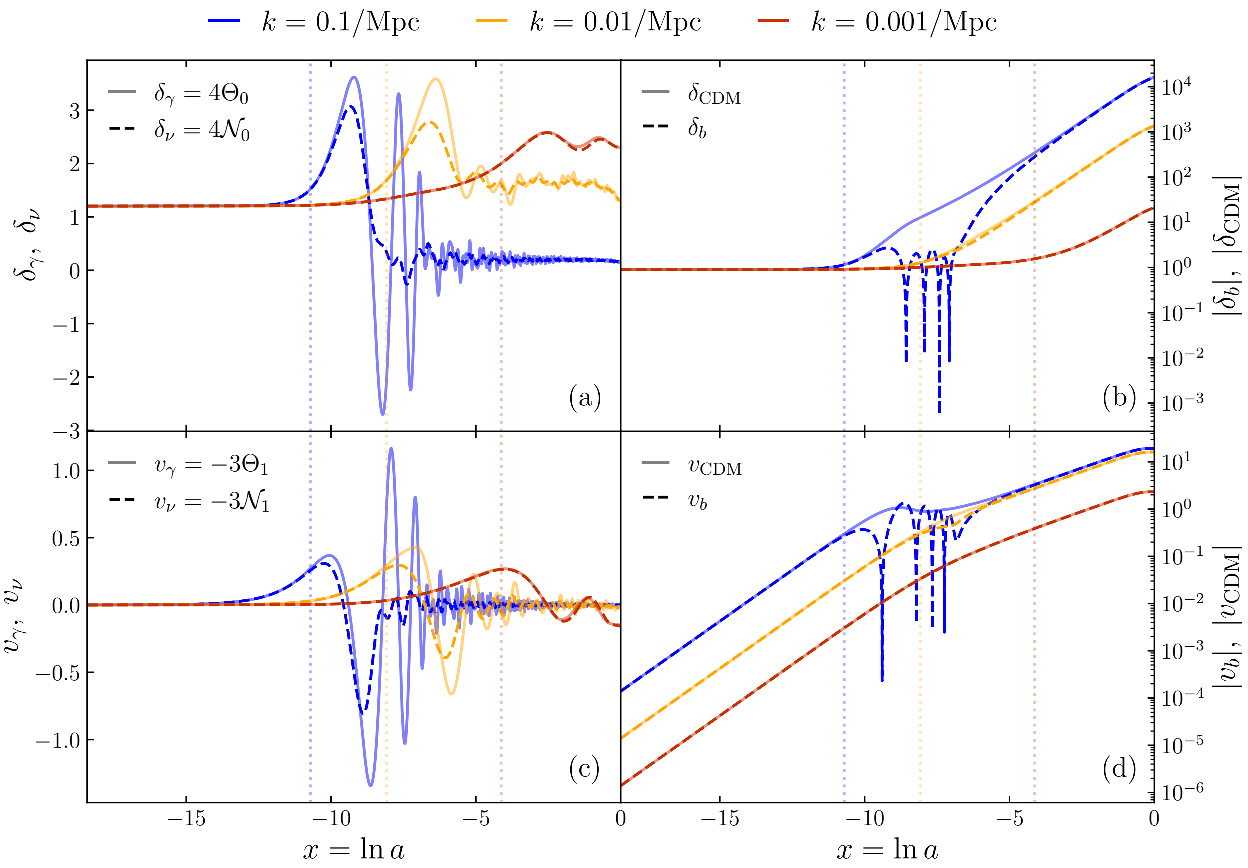

Fig. 12. (a) the photon and neutrino monopoles (density contrast), (b) the density contrast of CDM and baryons, (c) the dipole moments (bulk velocity) of the photon and neutrino perturbations, and (d) the bulk velocities of CDM and baryons, all as functions of the log-scale factor x. The colour of the lines indicate the perturbation scale k; blue is the smallest scale, orange in the middle, and red is the largest scale. The vertical dotted lines correspond to the time of horizon entry of the given pertubation scale k associated with the colour, i.e. $x_{\mathrm{HE}}(k)$ s.t. $k \sim \mathcal{H}(x_{\mathrm{HE}})/c$.

radiation pressure resists gravitational collapse and drives acoustic oscillations. At small scales the perturbations are damped by photon diffusion and decay over time, while at larger scales which enter the horizon closer to matter–radiation equality, the growing gravitional potential wells boost to the amplitude before the perturbations begin to oscillate. The bulk velocity perturbations $v_\gamma = -3\Theta_1$ and $v_\nu = -3\mathcal{N}_1$, shown in panel (c), mirrors the oscillatory behaviour of the density contrasts, with decaying amplitude over time.

The CDM and baryon density contrasts δ_{CDM} and δ_b are displayed in panel (b), and their associated bulk velocities v_{CDM} and v_b are shown in panel (d). After entering the horizon, the CDM collapses under gravity without any opposing pressure and forms so-called *dark matter halos* that grow as $\delta \propto \ln a = x$ in the radiation era and $\delta \propto a = e^x$ in the matter era (Meszaros 1974). The earlier the mode enters the horizon, the more time it has to grow, which is why we observe that the smaller scale perturbations reach the highest density contrasts of $\delta_{\mathrm{CDM}} \sim 10^4$.

Baryon perturbations that enter the horizon before recombination ($x_{\mathrm{HE}} < x_{\mathrm{rec}}$) initially begin to collapse under gravity, just as CDM does. However, since the baryons remain strongly coupled to the photons at this epoch, radiation pressure counteracts the gravitational infall and the two fluids oscillate together as a single photon-baryon plasma. These are the *baryon acoustic oscillations* (BAO), visible in panel (b) as oscillations between horizon entry and recombination for the smallest scale. The tight coupling between the photon and baryon fluids driving these oscillations is illustrated more clearly in Figure 13, where the two density contrasts are plotted together for $k = 0.1\,\mathrm{Mpc}^{-1}$. Upon decoupling, the baryons are released from radiation pressure and fall freely into the gravitational potential wells already set up by the collapsing CDM. The baryon density contrast then grows rapidly to catch up with that of the CDM, and the two track each other closely thereafter. Panel (b) therefore illustrates the necessity of CDM for structure formation: without the DM halos set up prior to recombination, the baryons at smaller scales could not have grown to the density contrasts observed today within the age of the Universe.

3.3.2. Metric perturbations and anisotropies sourced by neutrino and photon quadrupoles

The quadrupoles Θ_2/a^2 and $\mathcal{N}_2/a^2$ are shown in Figure 15. Both are non-zero on superhorizon scales, consistent with the adiabatic initial conditions from inflation, and both begin evolving after horizon entry. However, the neutrino quadrupole grows faster and to much larger amplitude, since it is decoupled and free-streaming in the early Universe, unimpeded by scattering. Thomson scattering in the photon-baryon plasma continuously

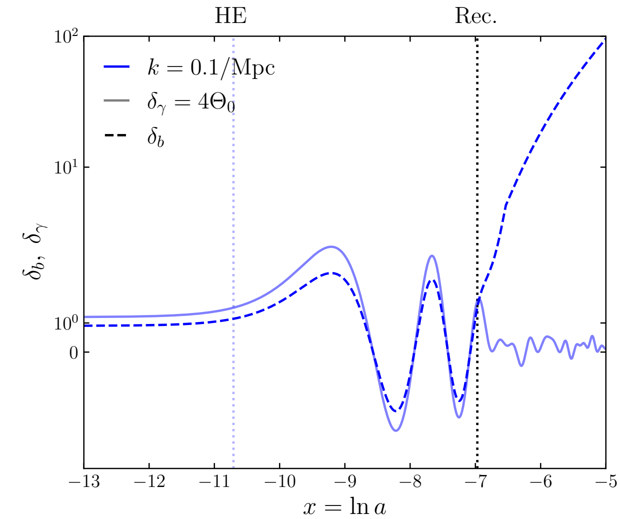

Fig. 13. The temporal evolution of the photon and baryon density contrasts, δ_γ and δ_b, at wavenumber $k = 0.1$/Mpc. Between horizon entry (HE) and recombination (Rec.), indicated with vertical dotted lines, the perturbations oscillate together in a photon-baryon plasma tightly coupled by Thomson scattering.

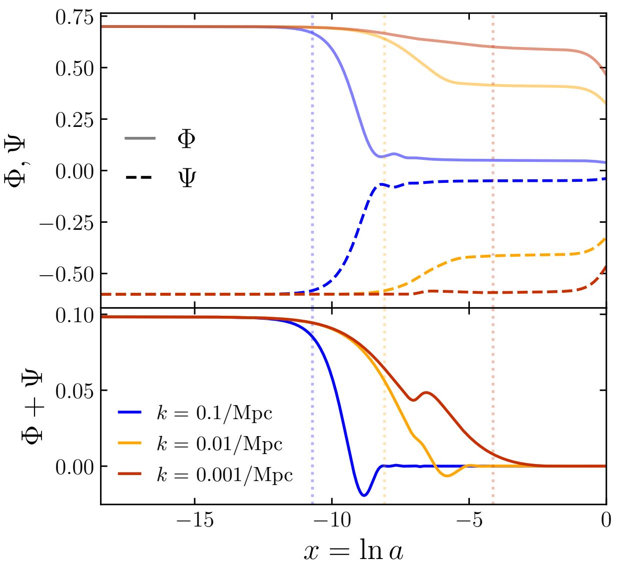

Fig. 14. Evolution of the scalar metric perturbations Φ and Ψ, and the anisotropic stress $\Phi + \Psi$. After the horizon entry of the perturbation modes (indicated by vertical dotted lines), the anisoptropic stress quickly decays and goes to zero.

suppresses Θ_2, keeping it small until recombination, after which it grows sharply as the mean free path increases. At late times Θ_2/a^2 and $\mathcal{N}_2/a^2$ both decay to zero and are diluted by cosmic expansion.

Figure 14 displays the evolution of the scalar metric perturbations Φ and Ψ, as well as the anisotropic stress $\Phi + \Psi$. On superhorizon scales the two potentials are nearly equal and opposite, so $\Phi + \Psi$ is small but non-zero. Even before horizon entry, the free-streaming neutrinos carry a considerable non-zero

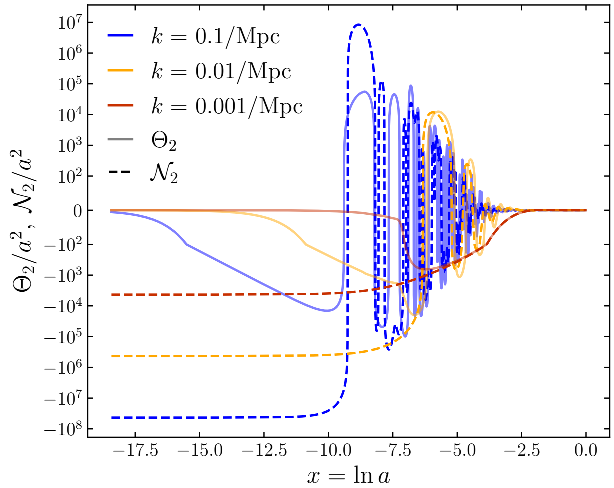

Fig. 15. The photon and neutrino quadrupole moments as functions of the log-scale factor x. The colour of the lines represent the perturbation scales.

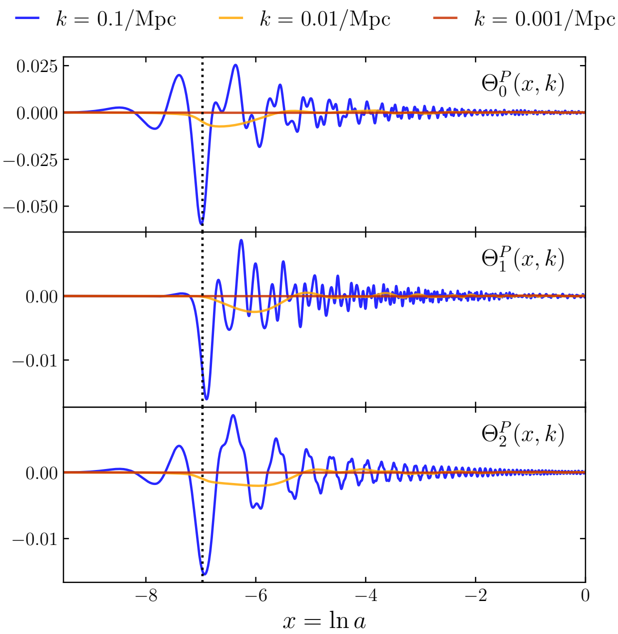

Fig. 16. The evolution of the first three photon polarisation multipoles ($\ell = 0, 1, 2$). Smaller-scale polarisation perturbations are sourced around the time of recombination (vertical black dotted line), while the largest scales are mostly unfazed. All the polarisation perturbations decay to zero in the late Universe.

quadrupole $\mathcal{N}_2$, as seen in Figure 15, which sources a small anisotropic stress through Eq.(24). After horizon entry, radiation pressure prevents the potentials from being sustained and they begin to decay and oscillate. As matter starts to dominate and the neutrino and photon quadrupoles are diluted, the anisotropic stress decays and $\Phi \approx -\Psi$ is restored at around $x \sim 2.5$.

3.3.3. Photon polarisation perturbations

The polarisation multipoles Θ_0^P, Θ_1^P, and Θ_2^P, seen in Figure 16, are completely suppressed before recombination, as frequent Thomson scatterings continuously isotropise the photon distribution. Around recombination, all modes grow sharply as the mean free path increases and scattering becomes inefficient, so polarisation is sourced during this transition. Smaller-k modes are effectively unfazed by the perturbations sourced during recombination since the polarisation perturbations have not had enough time to propagate to such large scales. Larger-k modes develop larger amplitudes and oscillate more rapidly, and the amplitude decreases with increasing ℓ as power is suppressed higher up in the multipole hierarchy. After recombination is completed the oscillations decay and all multipoles tend to zero for all k-modes. The polarisation perturbations present therefore a minor correction to the overall power spectrum, and the effect is mostly visible at small scales (large k).

4. The CMB and matter power spectra

Having evolved the perturbations, we are now in a position to compute the observable power spectra that connect our theoretical model to data. Comparing the predicted power spectra to observations from Planck 2018 (Aghanim et al. 2020) and large-scale structure surveys provides a test of the model. We give a synopsis of the theoretical framework for computing the CMB temperature, cross TE, EE polarisation and matter power spectra, describe the numerical implementation, and present the results.

4.1. Theory

In this section we outline the theory that allows us to produce the different power spectra obtained from the simulated perturbations, once again following Winther (2026a,e). The CMB temperature power spectrum C_ℓ is constructed from the photon transfer functions $\Theta_\ell(k)$, which we evaluate efficiently at high ℓ using the line-of-sight integral of Seljak & Zaldarriaga (1996). The same approach yields the polarisation spectra C_ℓ^{EE} and C_ℓ^{TE}, and we conclude with the matter power spectrum $\mathcal{P}(k)$.

4.1.1. CMB power spectrum

To get the CMB power spectrum from the perturbations we evolved in the previous section, we must decompose the CMB temperature field into an expansion in spherical harmonics (SH)

$$T(\hat{\boldsymbol{n}}) = \sum_{\ell m} a_{\ell m} Y_{\ell m}(\hat{\boldsymbol{n}}),$$

where $\hat{\boldsymbol{n}}$ is the direction on the sky, $a_{\ell m}$ are the spherical harmonic coefficients, and $Y_{\ell m}$ are the spherical harmonics. From the $a_{\ell m}$ coefficients we can evaluate the expectation value of their squares to get the CMB power spectrum

$$C_\ell \equiv \left\langle |a_{\ell m}|^2 \right\rangle = \langle a_{\ell m} a_{\ell m}^* \rangle. \tag{25}$$

It can be shown that the SH coefficients in terms of Θ are given by the integral

$$a_{\ell m}(\boldsymbol{x}, t) = \int d\Omega_{\hat{\boldsymbol{p}}} Y_{\ell m}^*(\hat{\boldsymbol{p}}) \Theta(\boldsymbol{x}, \hat{\boldsymbol{p}}, t).$$

We can then then plug in the expansion of Θ in multipoles of the Legendre polynomials (see Eq.(23)) and evaluate Eq.(25). This yields a rather messy integral that fortunately can be massaged into a more workable form

$$C_\ell = \frac{2}{\pi} \int k^2 \mathcal{P}_{\text{prim.}}(k) \Theta_\ell^2(k) dk.$$

Here Θ_ℓ are the photon multipoles and $\mathcal{P}_{\text{prim.}}(k)$ is the *primordial power spectrum*, which most inflation models predict to be a Harrison-Zel'dovich spectrum, given by

$$\frac{k^3}{2\pi^2} \mathcal{P}_{\text{prim.}}(k) = A_s \left(\frac{k}{k_{\text{pivot}}} \right)^{n_s - 1},$$

with $k_{\text{pivot}} = 0.05/\text{Mpc}$ is some wavenumber that yields a power spectrum amplitude $A_s \sim 2 \times 10^{-9}$, and $n_s \sim 0.96$ is the so-called *spectral index* of scalar perturbations. We therefore end up with the final expression for the CMB power spectrum

$$C_\ell = 4\pi \int_0^\infty A_s \left(\frac{k}{k_{\text{pivot}}} \right)^{n_s - 1} \Theta_\ell^2(k) \frac{dk}{k}, \tag{26}$$

which is the one we evaluate numerically from the perturbations we simulated in the previous section.

4.1.2. Line-of-sight integral

For us to compare our model to observational data, we need to evaluate the CMB power spectrum, Eq.(26), up to $\ell \sim 2000$, which requires knowing Θ_ℓ up to this order. As discussed in Section 3.2.2, we do not evaluate the multipoles directly in the ODE solver up to such high orders, since this would mean integrating thousands of coupled ODEs and would be computationally exhaustive. We instead solve the system up to $\ell_{\text{max}} = 8$ to ensure the first few multipoles are accurate, and then recover the higher-order multipoles using the line-of-sight (LOS) integral. This method was developed by Seljak and Zaldarriaga (Seljak & Zaldarriaga 1996) and allows us to evaluate the multipoles to much higher order using only the first few. Starting from the perturbed Boltzmann equation for photons,

$$\begin{aligned} &\frac{\partial \Theta}{\partial t} + \frac{ik\mu}{a} \Theta + \left(\frac{\partial \Phi}{\partial t} + \frac{ik\mu}{a} \Psi \right) \\ &\quad = n_e \sigma_T \left[\Theta_0 - \Theta + i\mu v_b - \frac{3\mu^2 - 1}{4} \Pi \right], \end{aligned}$$

one can derive the LOS integral giving the present-day photon multipoles (Winther 2026a),

$$\Theta_\ell(k, x = 0) = \int_{-\infty}^0 \tilde{S}(k, x) j_\ell[k(\eta_0 - \eta)] \mathrm{d}x.$$

Here j_ℓ are the spherical Bessel functions (see Appendix C), and $\tilde{S}$ is the source function, given by

$$\begin{aligned} \tilde{S}(k, x) = &\tilde{g} \left[\Theta_0 + \Psi + \frac{1}{4} \Pi \right] + e^{-\tau} \left[\Psi' - \Phi' \right] \\ &- \frac{1}{ck} \frac{\mathrm{d}}{\mathrm{d}x} (\mathcal{H} \tilde{g} v_b) + \frac{3}{4c^2 k^2} \frac{\mathrm{d}}{\mathrm{d}x} \left[\mathcal{H} \frac{\mathrm{d}}{\mathrm{d}x} (\mathcal{H} \tilde{g} \Pi) \right]. \end{aligned}$$

The first term in $\tilde{S}$ is the Sachs-Wolfe (SW) effect, arising from the uneven gravitational redshift of CMB photons at the surface of last scattering. The second is the integrated Sachs-Wolfe (ISW) term, which accounts for photons gaining or losing energy as they traverse decaying or growing potential wells along the line of sight. The third is the Doppler term, sourced by the bulk velocity of the photon-baryon plasma at last scattering. The last term is sourced by the polarisation potential Π, and is aptly referred to as the polarisation term.

The E-mode polarisation source function is given by

$$S_E(k, x) = \frac{3\tilde{g}\Pi}{4k^2(\eta_0 - \eta)^2},$$

which we use to evaluate the LOS integral and obtain $\Theta_\ell^E(k, x = 0)$, needed for computing the E-mode polarisation and cross power spectra C_ℓ^{EE} and C_ℓ^{TE}.

4.1.3. Matter power spectrum

Another observable quantity we can calculate is the matter power spectrum given by

$$\mathcal{P}(k, x) = |\Delta_M(k, x)|^2 \mathcal{P}_{\text{prim.}}(k),$$

where we define

$$\Delta_M(k, x) \equiv \frac{2}{3}\left(\frac{ck}{H_0}\right)^2 \frac{\Phi(k, x)}{\Omega_{M0} a^{-1}}.$$

The matter power spectrum $\mathcal{P}(k, x)$ is a measure of the amplitude of density perturbations as a function of scale. It is defined through the two-point correlation of the density contrast $\delta(\boldsymbol{k})$, and a larger $\mathcal{P}(k, x)$ at a given k indicates stronger clustering on that scale compared to a smooth density field.

4.2. Implementation details

4.2.1. Numerical LOS integration

The most computationally demanding step after solving the perturbation equations is the LOS integration. We therefore evaluate Θ_ℓ at 63 semi-evenly spaced multipoles up to $\ell = 2000$, listed in Table D.1, and spline the resulting power spectrum to obtain C_ℓ at all intermediate values.

To speed up the integration, we precompute splines of the spherical Bessel functions at each required ℓ rather than calling the GSL library at every integration step. The Bessel functions are sampled at linearly spaced arguments $y = k(\eta_0 - \eta)$ from zero to $y = k_{\max}\eta_0$ with a spacing of $\Delta y = \pi/8$, so that each period is sampled by 16 points and the rapid oscillations are accurately captured by the cubic spline.

The LOS integral is then evaluated using trapezoidal integration from $x_{\text{start}} = \ln(10^{-8}) \simeq -18.42$ to $x_{\text{end}} = 0$ for wavenumbers in the range $k_{\min} = 5 \times 10^{-5}\,\text{Mpc}^{-1}$ to $k_{\max} = 1.0\,\text{Mpc}^{-1}$. For each k, The LOS integral is evaluated numerically using the trapezoidal rule,

$$\Theta_\ell(k) = \sum_{i=1}^{N_x} \tilde{S}(x_i, k)\, j_\ell[k(\eta_0 - \eta(x_i))]\Delta x,$$

where $\Delta x = |x_{\text{end}} - x_{\text{start}}|/(N_x - 1)$ is the uniform spacing in x and $N_x = 700$. The wavenumbers are sampled linearly with a resolution of $\Delta k = \pi/4\eta_0$.

We follow the same LOS integration procedure for calculating the E-mode polarisation multipoles Θ_ℓ^E, only replacing $\tilde{S}$ with S_E.

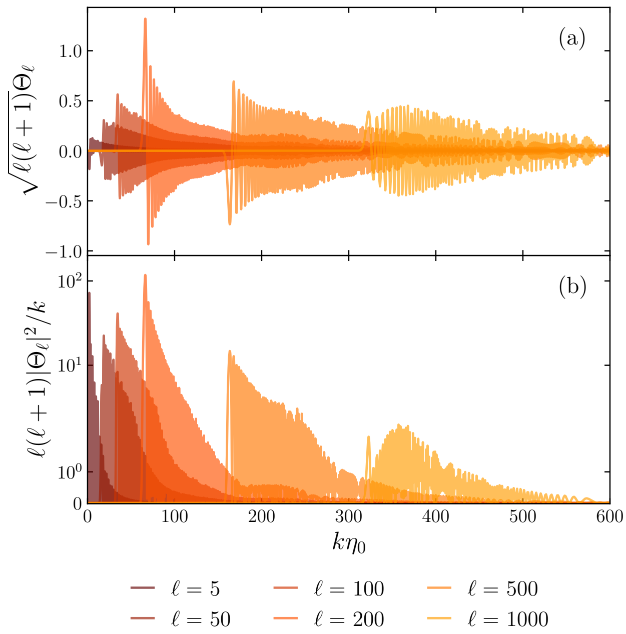

Fig. 17. (a) The transfer functions Θ_ℓ for six different ℓ, and (b) the power passed to the integrand of C_ℓ for the same six multipoles ℓ.

4.2.2. Producing the CMB power spectrum and map

Having solved for the Θ_ℓ and Θ_ℓ^E multipoles, we are now in the place to perform the integral in Eq.(26) and solve for the CMB power spectrum C_ℓ, as well as the cross and polarisation power spectra; C_ℓ^{TE} and C_ℓ^{EE}. This is done through the trapezoidal integral method, just as we did for the LOS integrals, sampling log-spaced k-values with resolution $\Delta(\ln k) = \pi/32k_{\max}\eta_0$,

$$C_\ell = 4\pi \sum_{i=1}^{N_k} \mathcal{P}_{\text{prim.}}(k_i)\Theta_\ell^2(k_i)\Delta(\ln k),$$

where $k_1 = k_{\min}$, $k_{N_k} = k_{\max}$, and

$$N_k = \frac{\ln(k_{\max}) - \ln(k_{\min})}{\Delta(\ln k)}.$$

The same is done for C_ℓ^{TE} and C_ℓ^{EE}, replacing one or both of the Θ_ℓ-factors in the integrand by Θ_ℓ^E. Then finally we spline the power spectra to get the values in between the ℓ-values listed in Table D.1.

In order to qualitatively compare our simulated CMB with observations, we produce the simulated CMB map using the Python library `healpy` (Zonca et al. 2019; Górski et al. 2005) from the computed power spectrum C_ℓ.

4.3. Results

4.3.1. Transfer functions

The temperature transfer functions Θ_ℓ are shown in Figure 17. Panel (a) displays the transfer functions $\sqrt{\ell(\ell+1)}\,\Theta_\ell$ as a function of $k\eta_0$, and a clear feature is that each multipole is localised around $k\eta_0 \sim \ell$, which follows directly from the spherical Bessel functions $j_\ell[k(\eta_0 - \eta)]$ peaking near their argument: each multipole ℓ is therefore sensitive to a characteristic physical scale.

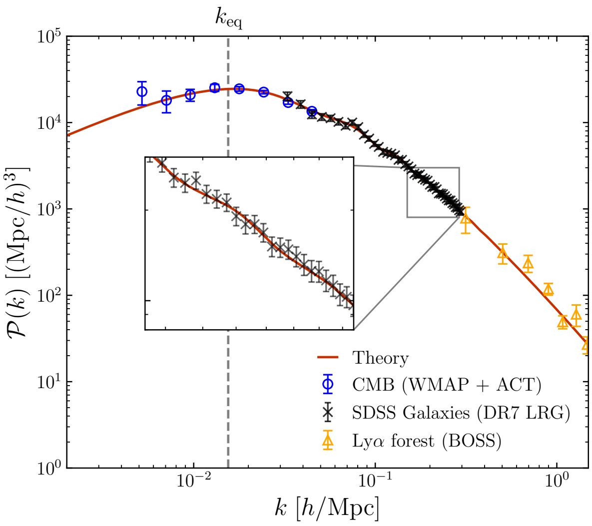

Fig. 18. The simulated theoretical current day matter power spectrum $\mathcal{P}(k, x = 0)$ along with observations from WMAP + ACT of the CMB; SDSS galaxy survey (DR7 LRG); and Lyα forest (BOSS). The vertical dashed line marks the location of the equality scale $k_{\text{eq}} \equiv \mathcal{H}(x_{\text{eq}}^{MR})/c$.

The oscillations within each transfer function reflects the acoustic oscillations in the photon-baryon plasma, with the number of oscillations increasing with ℓ, and the amplitude decreasing at large ℓ due to diffusion damping (Silk damping).

Panel (b) shows the power $\ell(\ell + 1)|\Theta_\ell|^2/k$, where the peaks shift to larger $k\eta_0$ with increasing ℓ, consistent with the $k\eta_0 \sim \ell$ scaling. The overall envelope decays at large $k\eta_0$ due to the exponential suppression from diffusion damping. The low-ℓ modes are an exception: these correspond to large angular scales that never entered the horizon during radiation domination and are therefore unaffected by acoustic oscillations and diffusion damping.

4.3.2. Matter power spectrum

The current day matter power spectrum $\mathcal{P}(k)$ derived from our model is shown in Figure 18, alongside observational data from three independent probes: CMB lensing (WMAP + ACT), galaxy clustering (SDSS DR7 LRG), and the Lyα forest (BOSS). The theoretical prediction is in excellent agreement with all three datasets.

The spectrum rises toward a peak near $k_{\text{eq}} \equiv \mathcal{H}(x_{\text{eq}}^{MR})/c$, the wavenumber corresponding to the mode that entered the horizon at matter-radiation equality, and falls off as a power law on either side of this turnover scale. Modes with $k < k_{\text{eq}}$ entered the horizon during matter domination and after recombination, such that they grew virtually unimpeded by the radiation pressure, preserving the near scale-invariant primordial spectrum $\mathcal{P}_{\text{prim.}}(k) \propto k^{n_s}$ on large scales. Modes with $k > k_{\text{eq}}$ entered the horizon during radiation domination, when radiation pressure prevented gravitational collapse and the growth of the CDM perturbations was logarithmically suppressed, the so-called Meszaros effect (Meszaros 1974). This suppression causes the characteristic turnover and the subsequent power-law decay at small scales. At the largest scales the Lyα and SDSS data overlap well with each other and with the theory, as shown in the inset, confirming that the model captures the correct shape of the power spectrum across a wide range of physical scales.

4.3.3. CMB and polarisation power spectra

The temperature power spectrum C_ℓ is shown in Figure 19, alongside Planck 2018 data (Aghanim et al. 2020) at both low and high ℓ. The theoretical prediction is in excellent agreement with the data across the full range of multipoles. The spectrum displays the characteristic series of acoustic peaks, arising from the oscillations of the photon-baryon plasma at recombination: modes that were at the extrema of their oscillation at last scattering contribute enhanced power, while those caught at a node are suppressed. The peaks are progressively damped at high ℓ by diffusion damping, which smooths out small-scale anisotropies. The lower panel decomposes the total power into its constituent source terms. The SW term dominates at large scales ($\ell \lesssim 100$), producing the characteristic plateau, while the Doppler and SW terms together make up the acoustic peaks at intermediate scales. The ISW contribution is subdominant across most of the spectrum but adds power at the largest scales through the decay of gravitational potentials. The polarisation term is the smallest contribution but becomes relatively more important at high ℓ. Also shown is a simplified model run with no helium, reionisation, neutrinos, nor polarisation. The acoustic peaks are visibly shifted to the left (towards lower ℓ) and has a strikingly higher third peak compared to the full model. The more pronounced high ℓ acoustic peaks illustrate how the extra components in the full theory suppress power at small scales.

The largest of the acoustic peaks is the first, located at around $\ell \sim 200$. This matches well with the characteristic scale set up by the BAO in the early Universe. In fact, we previously calculated the sound horizon at decoupling to be $r_s = 144.413$ Mpc, Eq.(21), which corresponds to a wavenumber of $k_s = 1/r_s \simeq 6.925 \times 10^{-3}$/Mpc, which leaves a signature in the power spectrum at $\ell \sim k\eta_0 \simeq 194.7$.

The TE cross power spectrum and the EE polarisation power spectrum are shown in Figure 20. Both are in excellent agreement with Planck 2018 data across the full multipole range. The TE spectrum oscillates between positive and negative values, reflecting the phase relationship between the temperature and polarisation fields; polarisation is generated at last scattering by the quadrupole anisotropy of the radiation field, which is $\pi/2$ out of phase with the temperature monopole, causing the cross spectrum to change sign at each acoustic peak. The EE spectrum displays a similar series of oscillations, with peaks offset relative to those in C_ℓ for the same reason, and is suppressed by roughly four orders of magnitude compared to the temperature spectrum, consistent with polarisation being a small correction sourced only near last scattering.

Finally, Figure 21 shows a simulated CMB temperature map generated from the computed C_ℓ. The map is a generated random realisation of the temperature perturbations and displays temperature fluctuations of order $\pm 500\,\mu$K. The map is consistent with a Gaussian random field, as expected from inflation, and is a qualitative visual confirmation that the computed power spectrum reflects the correct appearance of the CMB as the observations, e.g. the WMAP CMB map (see Figure 1).

5. Conclusions

In this project we have built a numerical Einstein-Boltzmann code that simulates cosmic structure formation in a flat ΛCDM universe and computes the CMB and matter power spectra from first principles. Starting from the background cosmology, we established the expansion history and tested the model against SNIa luminosity distance data, recovering the well-known Hub-

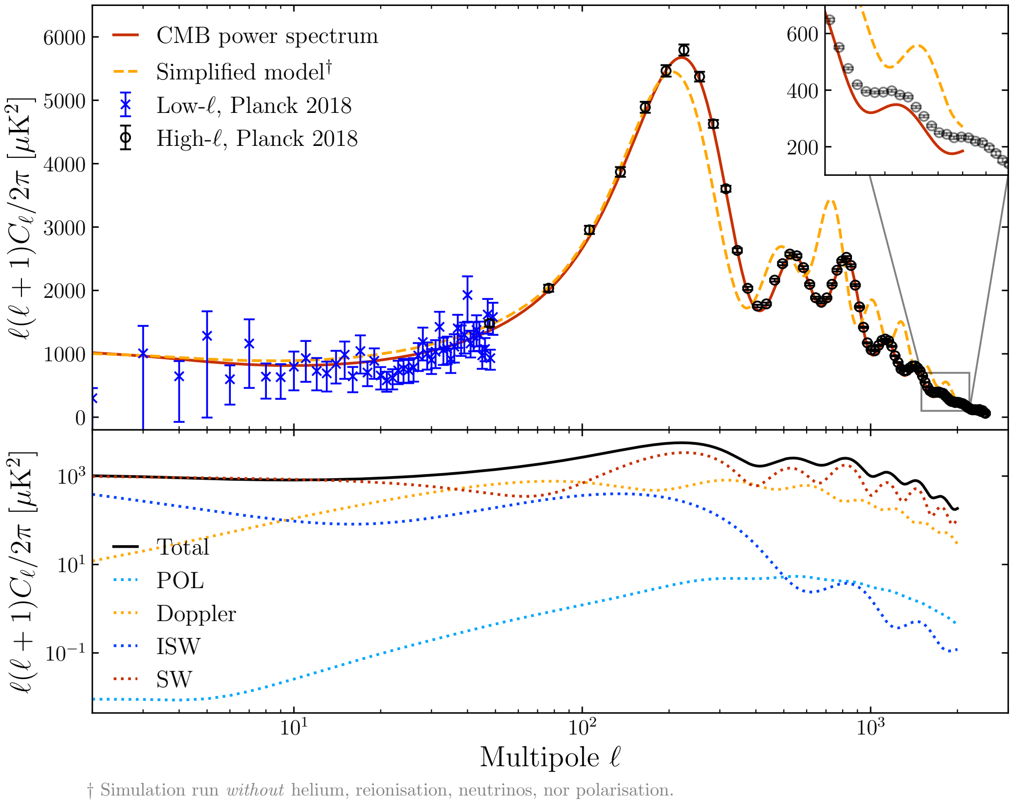

Fig. 19. The top panel shows the power spectra for the theoretical CMB and a simplified model (for which there is no helium nor neutrinos, and reionisation and polarisation has been turned off) together with observational data from Planck 2018 (Aghanim et al. 2020). The lower panel breaks down the total power spectrum displaying the power spectra resulting from turning on each of the four terms in the source function $\tilde{S}$ separately while the others are turned off; Sachs-Wolfe (SW), integrated Sachs-Wolfe (ISW), Doppler, and polarisation (POL).

ble tension between the CMB-inferred and late-universe galaxy survey values of h. We then computed the recombination history, solving for the free electron fraction through the Saha and Peebles regimes and including helium and reionisation, and determined the sound horizon at decoupling to be $r_s \simeq 144.413\,\mathrm{Mpc}$, in good agreement with Planck 2018. The coupled perturbation equations for CDM, baryons, photons, neutrinos, and the gravitational potentials were then evolved from adiabatic inflationary initial conditions, using the tight-coupling approximation to handle the stiff pre-recombination system. Finally, the photon transfer functions were projected onto multipoles via the line-of-sight integral, and the resulting CMB temperature, polarisation, and cross power spectra as well as the matter power spectrum were compared to observational data, finding excellent agreement across all observables.

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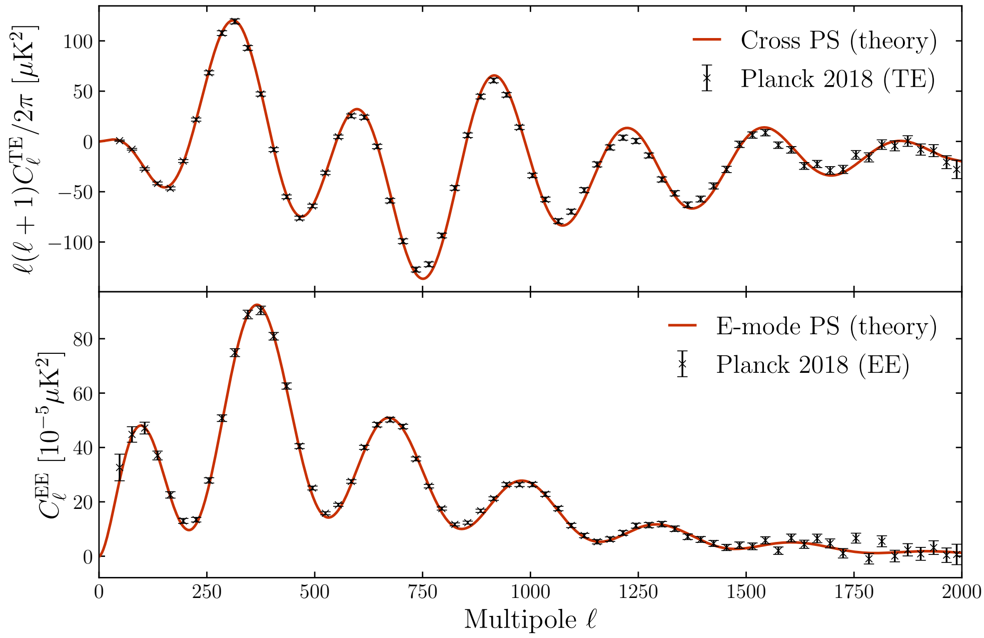

Fig. 20. The cross power spectrum C_ℓ^{TE} and the E-mode polarisation power spectrum C_ℓ^{EE} plotted against observational data from Planck 2018 (Aghanim et al. 2020).

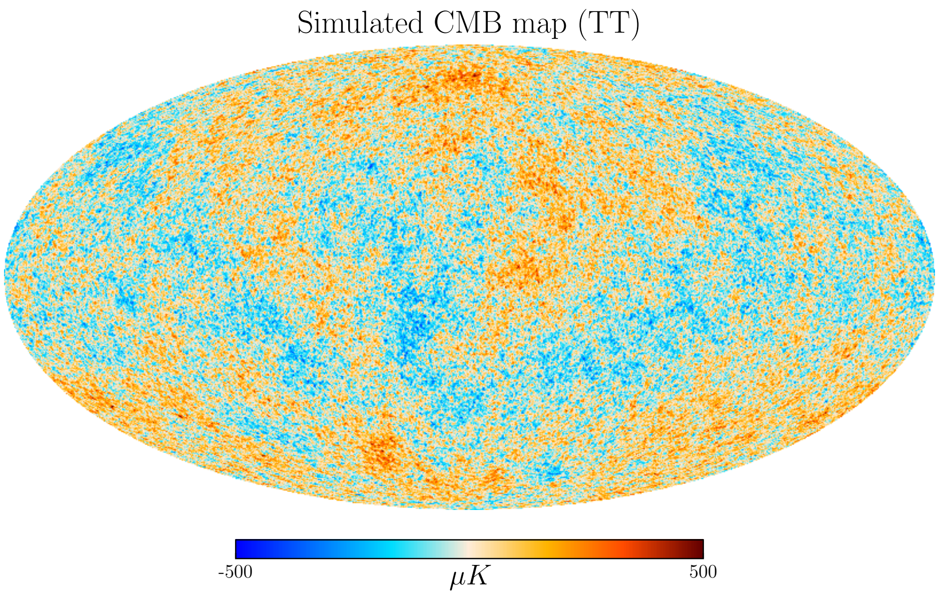

Fig. 21. CMB map produced from the simulated power spectrum. Generated using the Python library `healpy` (Zonca et al. 2019; Górski et al. 2005). Colourmap taken from Zonca (2013).

Appendix A: Derivatives of the conformal Hubble function

The Hubble function is given by

$$H = H_0\sqrt{\frac{\Omega_{b0}+\Omega_{\mathrm{CDM}0}}{a^3}+\frac{\Omega_{\gamma 0}+\Omega_{\nu 0}}{a^4}+\frac{\Omega_{k0}}{a^2}+\Omega_{\Lambda 0}},$$

where we can define $\Omega_{M0} \equiv \Omega_{b0}+\Omega_{\mathrm{CDM}0}$ and $\Omega_{R0} \equiv \Omega_{\gamma 0}+\Omega_{\nu 0}$. The first derivative of the conformal Hubble function $\mathcal{H} \equiv aH(a)$ with respect to $x \equiv \ln a$ is

$$\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} = \frac{\mathrm{d}a}{\mathrm{d}x}H + a\frac{\mathrm{d}H}{\mathrm{d}x} = \mathcal{H} + a^2\frac{\mathrm{d}H}{\mathrm{d}a}.$$

Let us compute the derivative of $H(a)$:

$$\frac{\mathrm{d}H}{\mathrm{d}a} = -\frac{H_0^2}{2H}\Big(3\Omega_{M0}a^{-4}+4\Omega_{R0}a^{-5}+2\Omega_{k0}a^{-3}\Big).$$

This gives

$$\begin{aligned}\frac{\mathrm{d}\mathcal{H}}{\mathrm{d}x} &= \mathcal{H} - a^2\frac{H_0^2}{2H}\Big(3\Omega_{M0}a^{-4}+4\Omega_{R0}a^{-5}+2\Omega_{k0}a^{-3}\Big)\\ &= \mathcal{H}\Big[1-\frac{3}{2}\Omega_M - 2\Omega_R - \Omega_K\Big].\end{aligned}$$

Here the Ω_i's refer to the density parameters as functions of scale factor, i.e. $\Omega_i(a)$, which should be evaluated at $a = e^x$.

The second derivative is a bit more cumbersome to calculate; from the product rule it follows that

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} &= \mathcal{H}\Big[1-\frac{3}{2}\Omega_m - 2\Omega_r - \Omega_K\Big]^2\\ &\quad + \mathcal{H}\frac{\mathrm{d}}{\mathrm{d}x}\Big[1-\frac{3}{2}\Omega_m - 2\Omega_r - \Omega_K\Big].\end{aligned}$$

Let B be the expression inside the big bracket. We thus need $\mathrm{d}B/\mathrm{d}x$, which is most easily calculated by factoring out $H_0^2/H(a)^2$ from the last three terms of B;

$$\begin{aligned}\frac{\mathrm{d}B}{\mathrm{d}x} &= -\frac{a}{2}\frac{\mathrm{d}}{\mathrm{d}a}\left(\frac{H_0^2}{H^2}\right)\Big(3\Omega_{M0}a^{-3}+4\Omega_{R0}a^{-4}+2\Omega_{K0}a^{-2}\Big)\\ &\quad -\frac{a}{2}\frac{H_0^2}{H^2}\frac{\mathrm{d}}{\mathrm{d}a}\Big(3\Omega_{M0}a^{-3}+4\Omega_{R0}a^{-4}+2\Omega_{K0}a^{-2}\Big)\\ &= -\frac{1}{2}\Big(3\Omega_M+4\Omega_R+2\Omega_K\Big)^2+\frac{1}{2}\Big(9\Omega_M+16\Omega_R+4\Omega_K\Big).\end{aligned}$$

From this we end up with second derivative

$$\begin{aligned}\frac{\mathrm{d}^2\mathcal{H}}{\mathrm{d}x^2} = \mathcal{H}\bigg\{&\Big[1-\frac{3}{2}\Omega_M-2\Omega_R-\Omega_K\Big]^2\\ &-\frac{1}{2}\Big(3\Omega_M+4\Omega_R+2\Omega_K\Big)^2+\frac{1}{2}\Big(9\Omega_M+16\Omega_R+4\Omega_K\Big)\bigg\}\bigg|_{a=e^x}\end{aligned}$$

Appendix B: Logarithmic scale factor at different cosmological events

Matter-radiation equality happens when $\Omega_M(a_{\mathrm{eq}}^{MR}) = \Omega_R(a_{\mathrm{eq}}^{MR})$. Using the expressions given in Eq.(6), we find

$$\frac{\Omega_{M0}}{(a_{\mathrm{eq}}^{MR})^3} = \frac{\Omega_{R0}}{(a_{\mathrm{eq}}^{MR})^4} \;\Rightarrow\; x_{\mathrm{eq}}^{MR} \equiv \ln a_{\mathrm{eq}}^{MR} = \ln\left(\frac{\Omega_{R0}}{\Omega_{M0}}\right).$$

In a similar way, we derive that the matter-dark energy equality occurs for

$$\frac{\Omega_{M0}}{(a_{\mathrm{eq}}^{\Lambda M})^3} = \Omega_{\Lambda 0} \;\Rightarrow\; x_{\mathrm{eq}}^{\Lambda M} \equiv \ln a_{\mathrm{eq}}^{\Lambda M} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{\Omega_{\Lambda 0}}\right).$$

The onset of cosmic acceleration is defined to be when $\ddot{a} = 0$. From the second Friedmann equation, Eq.(4), we get that this occurs whenever

$$\sum_i \rho_i(a_{\mathrm{acc}})(1+3w_i) = 0.$$

The relevant equations of state are $w_M = 0$, $w_R = 1/3$ and $w_\Lambda = -1$. We neglect the curvature parameter when working with fiducial cosmological parameter, and thus the condition for cosmic acceleration can be recast into

$$\begin{aligned}&\Omega_M + 2\Omega_R - 2\Omega_\Lambda = 0\\ \Rightarrow\quad &\frac{\Omega_{M0}}{(a_{\mathrm{acc}})^3}+\frac{2\Omega_{R0}}{(a_{\mathrm{acc}})^4} - 2\Omega_{\Lambda 0} = 0.\end{aligned}$$

At late times, the radiation density parameter can be neglected, so we end up with

$$x_{\mathrm{acc}} \equiv \ln a_{\mathrm{acc}} = \frac{1}{3}\ln\left(\frac{\Omega_{M0}}{2\Omega_{\Lambda 0}}\right).$$

Appendix C: Legendre polynomials and spherical Bessel functions

The Legendre polynomials $P_\ell(\mu)$ are functions of $\mu = \cos\theta$, where θ is the angle between two vectors. They arise naturally in problems with spherical symmetry, such as expansions of gravitational or electrostatic potentials. The first four are

$$\begin{aligned}P_0(\mu) &= 1,\\ P_1(\mu) &= \mu,\\ P_2(\mu) &= \frac{3\mu^2-1}{2},\\ P_3(\mu) &= \frac{5\mu^3-3\mu}{2},\end{aligned}$$

and all higher-order polynomials can be generated via Bonnet's recursion formula,

$$\mu P_\ell(\mu) = \frac{\ell+1}{2\ell+1}P_{\ell+1}(\mu) + \frac{\ell}{2\ell+1}P_{\ell-1}(\mu).$$

The set $\{P_\ell\}_{\ell=0}^{\infty}$ forms an orthogonal basis for the space of square-integrable functions on $[-1, 1]$, with the orthogonality relation

$$\int_{-1}^{1} P_\ell(\mu)P_{\ell'}(\mu)\,\mathrm{d}\mu = \frac{2}{2\ell+1}\delta_{\ell\ell'}.$$

For our interest, this means that any sufficiently well-behaved function f can therefore be uniquely decomposed as a multipole expansion,

$$f\Big(\mu;\{x_i\}_{i=1}^N\Big) = \sum_{\ell=0}^{\infty}\frac{2\ell+1}{i^\ell}f_\ell\Big(\{x_i\}_{i=1}^N\Big)P_\ell(\mu),$$

where the multipole coefficients are extracted via the orthogonality relation,

$$f_\ell\Big(\{x_i\}_{i=1}^N\Big) \equiv \frac{i^\ell}{2}\int_{-1}^{1}P_\ell(\mu)\,f\Big(\mu;\{x_i\}_{i=1}^N\Big)\,\mathrm{d}\mu.$$

The spherical Bessel functions $j_\ell(x)$ are the radial counterparts to the Legendre polynomials: while the P_ℓ's describe the angular dependence of a system, the j_ℓ's encode the radial dependence. The first few are

$$j_0(x) = \frac{\sin x}{x},$$
$$j_1(x) = \frac{\sin x}{x^2} - \frac{\cos x}{x},$$
$$j_2(x) = \left(\frac{3}{x^3} - \frac{1}{x}\right)\sin x - \frac{3\cos x}{x^2},$$

and higher orders follow from the recursion relation

$$j_{\ell+1}(x) = \frac{2\ell+1}{x}\, j_\ell(x) - j_{\ell-1}(x).$$

A key identity linking the Bessel functions and the Legendre polynomials is the plane-wave expansion,

$$e^{i\mathbf{k}\cdot\mathbf{r}} = \sum_{\ell=0}^{\infty} i^\ell (2\ell+1)\, j_\ell(kr)\, P_\ell(\hat{k}\cdot\hat{r}),$$

which decomposes a plane wave into spherical waves of definite angular momentum ℓ, and explains the appearance of the i^ℓ factor in the multipole expansion above.

Appendix D: ℓ-values

Evaluated ℓ-values						
2	3	4	5	6	7	8
10	12	15	20	25	30	40
50	60	70	80	90	100	120
140	160	180	200	225	250	275
300	350	400	450	500	550	600
650	700	750	800	850	900	950
1000	1050	1100	1150	1200	1250	1300
1350	1400	1450	1500	1550	1600	1650
1700	1750	1800	1850	1900	1950	2000

Table D.1. The 63 ℓ-values used in the evaluation of the CMB power spectra.